

## Unit 10

# D.C. CIRCUITS

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54 items: 21 from official papers (132 marks), 33 from the booklet

### **Syllabus sections in this unit**

- 10.1 Practical circuits
- 10.2 Kirchhoff's laws
- 10.3 Potential dividers

## Unit 10: D.C. circuits

### 1. M/J 26/P22/Q5/a,b(i),c

also Unit 9 (2 marks); this unit 8 marks

- 5 (a) Three resistors of resistances  $R_1$ ,  $R_2$  and  $R_3$  are connected in a network in parallel as shown in Figure 5.1.

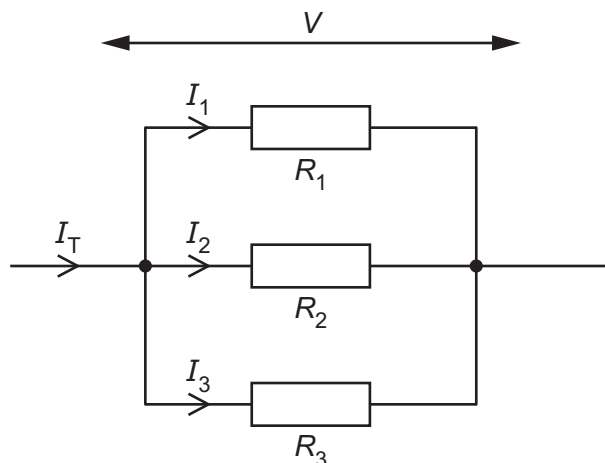


Figure 5.1

The currents in the resistors are  $I_1$ ,  $I_2$  and  $I_3$  respectively. The total current entering the network is  $I_T$ . The potential difference (p.d.) across the resistors is  $V$ .

Derive the equation

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

where  $R_T$  is the total resistance of the network.

[2]

- (b) Two identical resistance wires, each of resistance  $2.8\Omega$ , are connected in parallel to a battery with negligible internal resistance and electromotive force (e.m.f.)  $12\text{V}$  as shown in Figure 5.2.

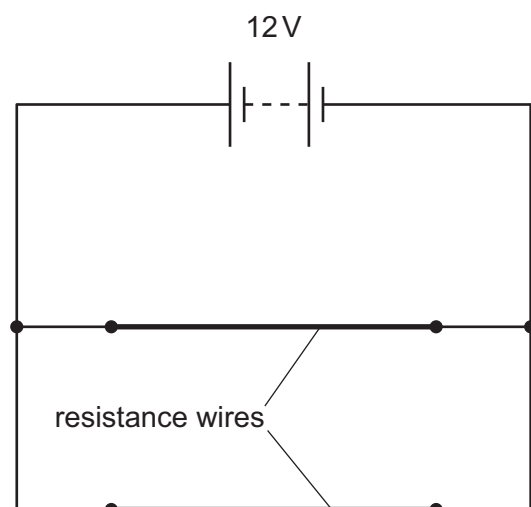


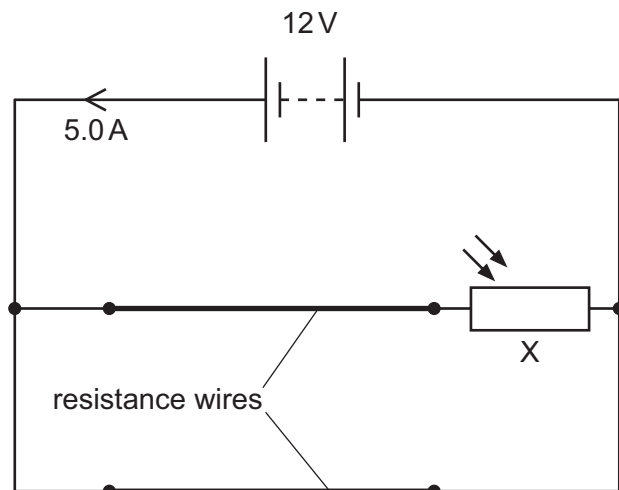
Figure 5.2

- (i) Calculate the current in the battery.

current =

A [2]

- (c) A component X is added to the circuit in **5(b)** as shown in Figure 5.3.



**Figure 5.3**

The current in the battery is now 5.0 A.

The resistance of each resistance wire remains  $2.8\ \Omega$ .

- (i) State the name of component X.

[1]

- (ii) Determine the resistance of component X.

resistance =

$\Omega$  [3]

- (iii) A change in the environment of component X causes the current in the battery to decrease. The resistance of the resistance wires remains constant.

State and explain how the environment of X has changed.

[2]

**2. M/J 26/P23/Q4/a,b(iii)**

- 4 A light-dependent resistor (LDR) is connected across the terminals of a cell that has internal resistance. A voltmeter is connected across the LDR.

- (a) Draw a circuit diagram to show this arrangement.

[3]

- (b) The cell has an electromotive force (e.m.f.) of 4.90 V.

At a particular light intensity, the LDR has a resistance of  $230\ \Omega$  and a potential difference (p.d.) across it of 4.78 V.

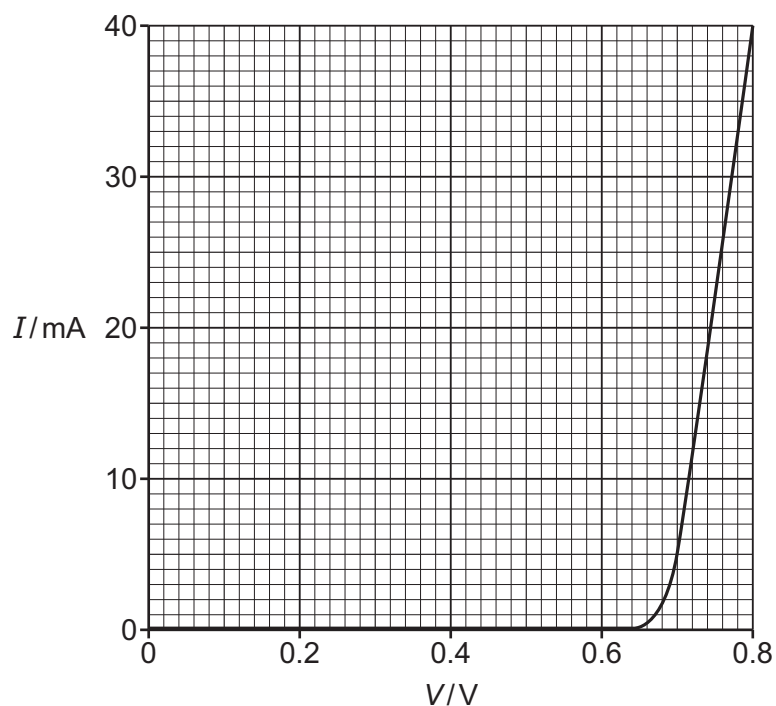
- (iii) Determine the internal resistance of the cell.

internal resistance =

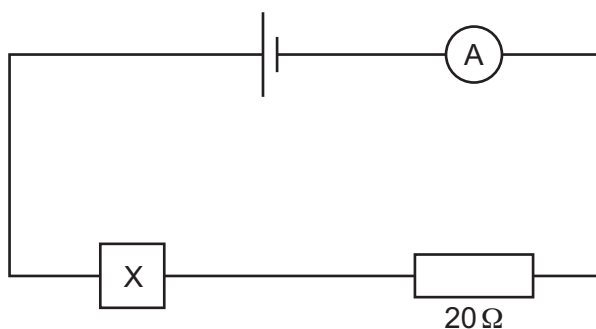
$\Omega$  [2]

**3. M/J 26/P24/Q5/c,e**

- 5 The  $I$ - $V$  characteristic of an electrical component X is shown in Figure 5.1.

**Figure 5.1**

- (c) Component X is connected in a circuit with a resistor of resistance  $20\ \Omega$  as shown in Figure 5.2.

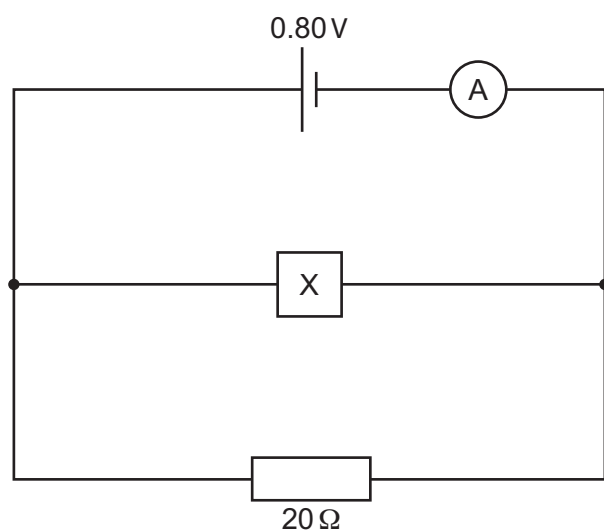
**Figure 5.2**

The current in the ammeter is  $5.0\ \text{mA}$  and the cell has negligible internal resistance.

Show that the electromotive force (e.m.f.) of the cell is  $0.80\ \text{V}$ .

[3]

- (e) The resistor of resistance  $20\ \Omega$  and component X are now connected in parallel to the cell in 5(c) in the circuit shown in Figure 5.3.

**Figure 5.3**

The current in component X is **not** negligible.

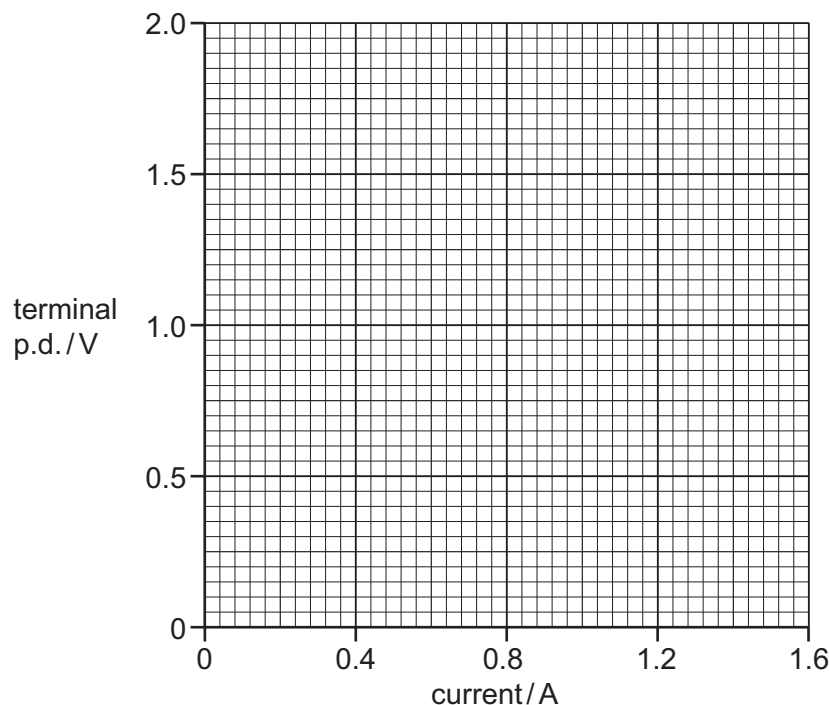
Use Figure 5.1 to determine the current in the ammeter.

current = \_\_\_\_\_ A [3]

**4. MAR 26/P22/Q6/b,c(ii)**

(b) A cell has electromotive force (e.m.f.) 1.50 V and internal resistance 0.75  $\Omega$ .

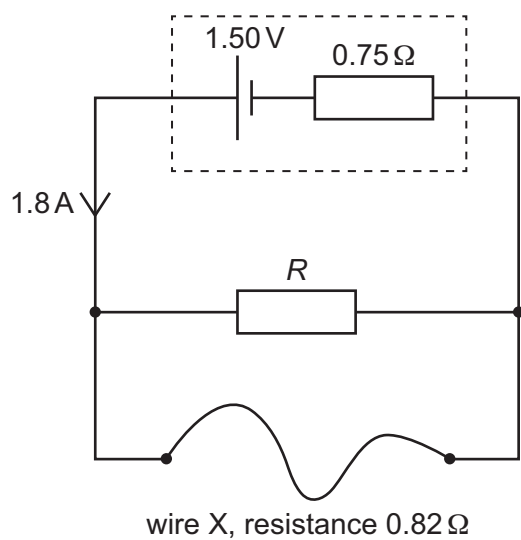
On Figure 6.1, draw the variation of the terminal potential difference (p.d.) with current in the cell for current between 0 A and 1.6 A.

**Figure 6.1**

[2]

(c) A metal wire X has length 6.8 m and cross-sectional area  $2.0 \times 10^{-7} \text{ m}^2$ . The resistance of the wire is 0.82  $\Omega$ .

The wire is connected to a circuit containing the cell in **6(b)** and a fixed resistor of resistance  $R$ , as shown in Figure 6.2.

**Figure 6.2**

The current in the cell is 1.8 A.

(ii) Calculate  $R$ .

$R =$

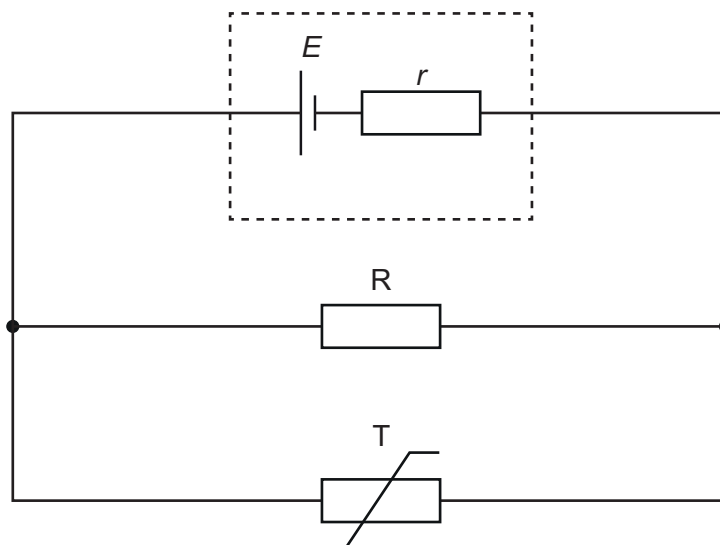
Ω [3]

**5. O/N 25/P21/Q5/a,b(ii),c**

- 5 (a) State Kirchhoff's first law.

[1]

- (b) Fig. 5.1 shows a circuit containing a thermistor  $T$  that has a negative temperature coefficient.

**Fig. 5.1**

- (ii) With reference to the current in the cell, explain why the current in **resistor R** decreases with increasing temperature of the thermistor.

[3]

- (c) The electromotive force (e.m.f.)  $E$  of the cell in Fig. 5.1 is 1.50 V. The internal resistance  $r$  of the cell is  $0.12\ \Omega$ .

Resistor  $R$  has a resistance of  $6.00\ \Omega$ .

At a particular temperature of the thermistor, the current in  $R$  is  $0.200\text{ A}$ .

For this temperature of the thermistor, determine:

- (i) the current in the cell

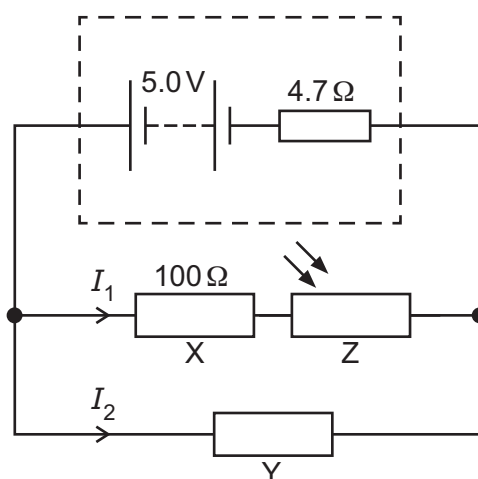
current = A [2]

- (ii) the resistance of the thermistor.

resistance =  $\Omega$  [2]

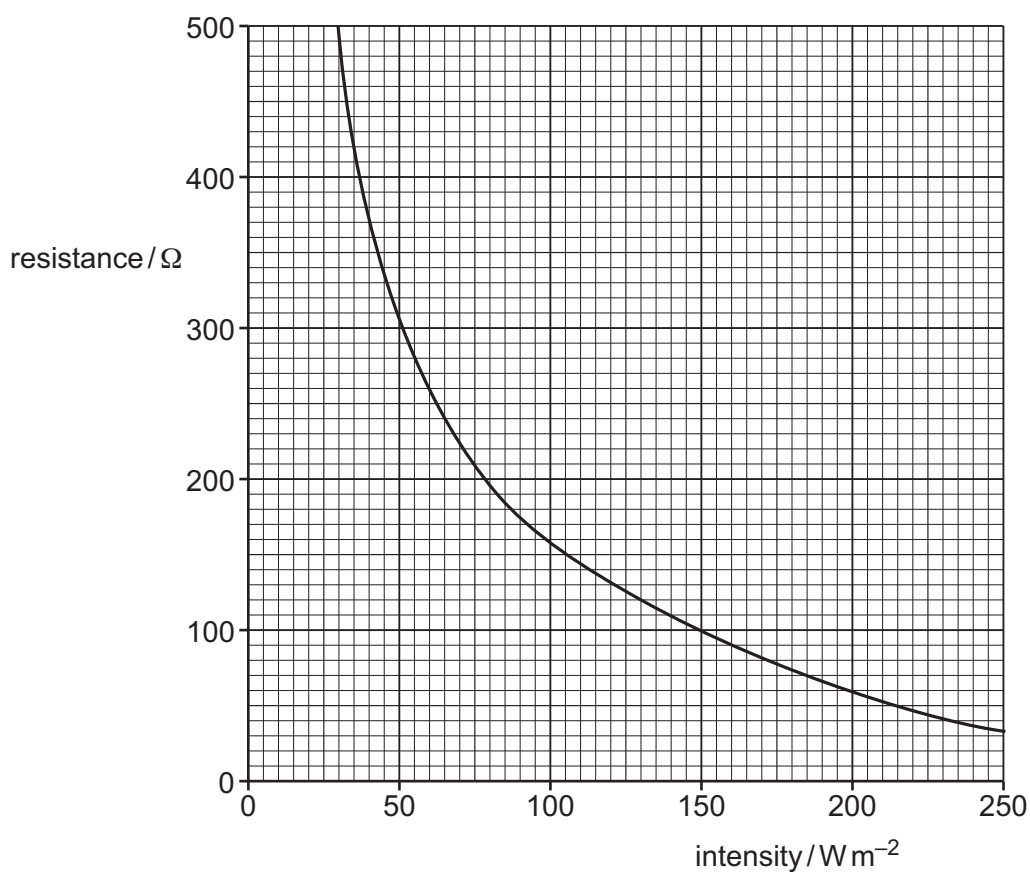
**6. O/N 25/P22/Q5/a,b(i,ii,iv)**

- 5 Fig. 5.1 shows a circuit containing a battery, two fixed resistors  $X$  and  $Y$ , and a light-dependent resistor (LDR)  $Z$ .

**Fig. 5.1**

The battery has electromotive force (e.m.f.) 5.0 V and internal resistance 4.7 Ω. The current in X is  $I_1$  and the current in Y is  $I_2$ .

The resistance of X is 100 Ω. The resistance of Z varies with the intensity of light incident on it as shown in Fig. 5.2.

**Fig. 5.2**

(a) State Kirchhoff's first law.

[1]

(b) The intensity of light incident on Z is  $130 \text{ W m}^{-2}$ . The current in the battery is 38 mA.

(i) Show that the terminal potential difference of the battery is 4.8 V.

[2]

- (ii) Calculate the current  $I_2$  in Y.

$$I_2 = \text{A [3]}$$

- (iv) The intensity of the light incident on Z decreases.

State and explain the effect on the terminal potential difference of the battery.

[3]

**7. O/N 25/P24/Q5/a(ii)**

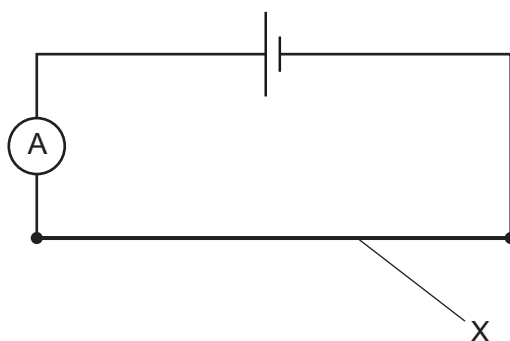
- 5** A student uses a circuit containing an ammeter, a voltmeter and a cell to take measurements to determine the resistance of a length of nichrome wire.

- (ii) Draw a circuit diagram to show how the components should be connected. Use the symbol for a resistor to represent the nichrome wire.

[2]

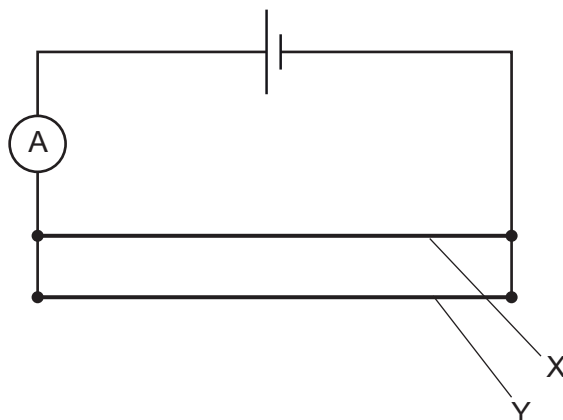
**8. M/J 25/P21/Q6/b(i)**

- 6 A nichrome wire X of length 45 cm and cross-sectional area  $4.7 \times 10^{-7} \text{ m}^2$  is connected into the circuit shown in Fig. 6.1.

**Fig. 6.1**

The resistance of X is  $1.1 \Omega$ . The cell has electromotive force (e.m.f.) 1.3 V and negligible internal resistance.

- (b) Wire Y is identical to wire X. Wire Y is added to the circuit in parallel with wire X, as shown in Fig. 6.2.

**Fig. 6.2**

State and explain the effect, if any, of this change on:

- (i) the reading on the ammeter

[2]

**9. M/J 25/P22/Q6/b(ii,iii,iv),c**

- (b) A circuit contains four resistors and a battery of electromotive force (e.m.f.) 8.0 V with negligible internal resistance. When the variable resistor has resistance  $R$ , the currents in the circuit are 0.030 A,  $I_1$  and  $I_2$ , as shown in Fig. 6.1.

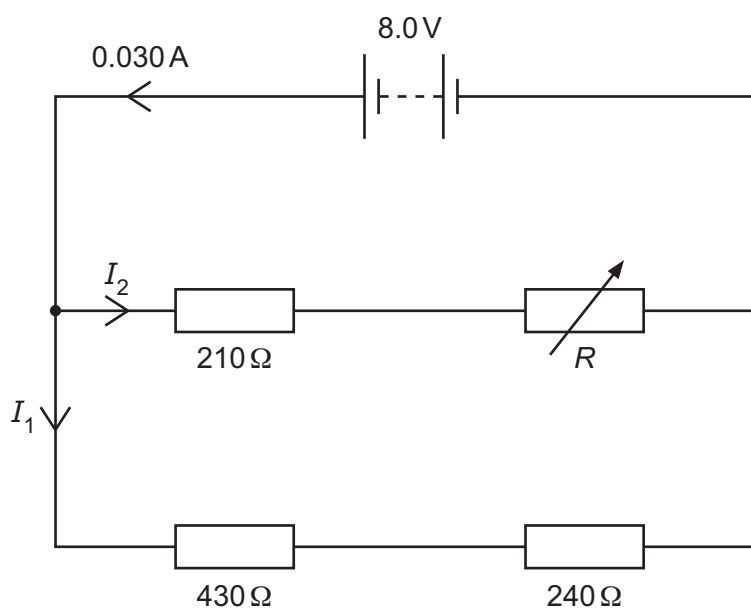


Fig. 6.1

(ii) Calculate  $I_1$ .

$I_1 =$  A [2]

(iii) Calculate  $I_2$ .

$I_2 =$  A [1]

(iv) Determine  $R$ .

$R =$  Ω [2]

(c) The variable resistor in (b) is fitted with a scale so that its resistance can be accurately determined.

The resistor of resistance  $240\ \Omega$  is now replaced by a new resistor X of unknown resistance. A galvanometer is connected as shown in Fig. 6.2.

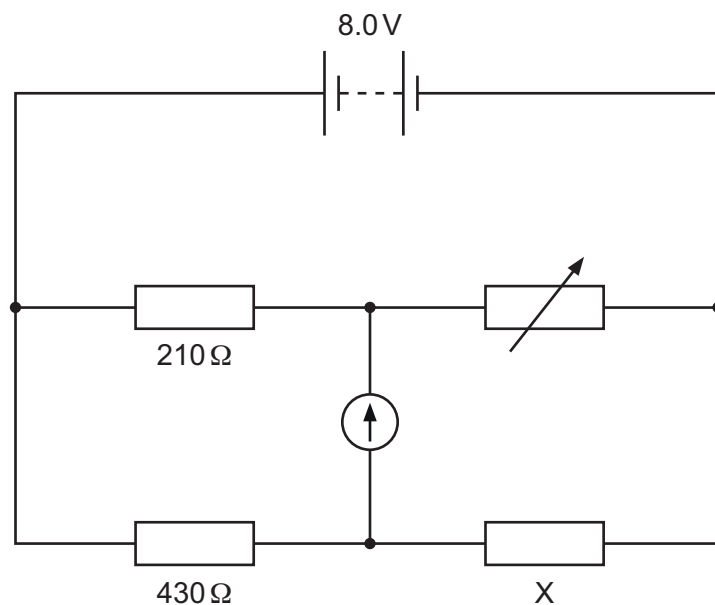


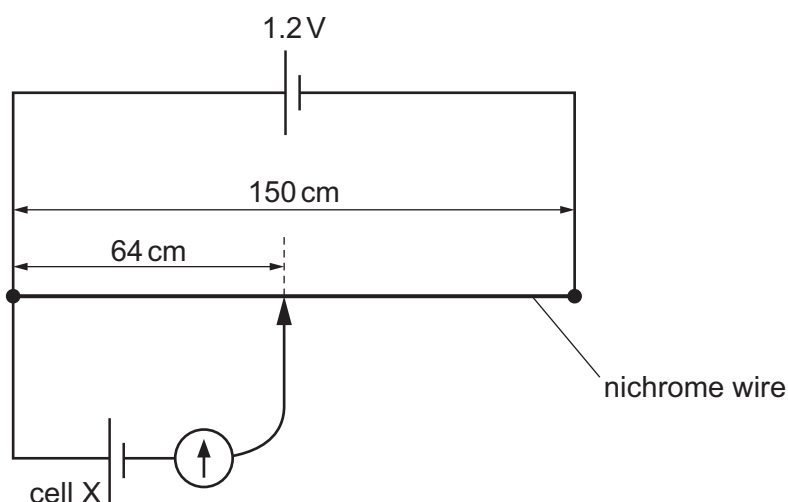
Fig. 6.2

With reference to ratios of resistances, explain how this circuit can be used to determine the resistance of X.

[2]

**10. M/J 25/P23/Q7/b**

- 7 A nichrome resistance wire has length 150 cm, cross-sectional area  $2.45 \times 10^{-7} \text{ m}^2$  and resistivity  $1.12 \times 10^{-6} \Omega \text{ m}$ .
- (b) The nichrome wire forms part of a potentiometer circuit together with a cell of electromotive force (e.m.f.) 1.2 V and negligible internal resistance, as shown in Fig. 7.1.

**Fig. 7.1** (not to scale)

The circuit is used to determine the e.m.f. of cell X.

The galvanometer is used in a null method to find the null point 64 cm from the left-hand end of the nichrome wire.

- (i) Explain what is meant by a null method.

[1]

- (ii) Calculate the e.m.f. of cell X.

e.m.f. = \_\_\_\_\_ V [2]

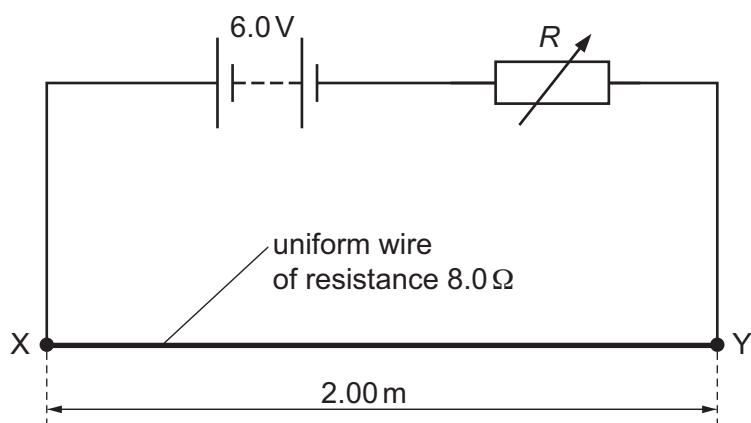
- (iii) The cell of e.m.f. 1.2 V is replaced by a new cell with the same e.m.f. but with an internal resistance that is **not** negligible.

State and explain the effect, if any, of the internal resistance of the new cell on the position of the null point.

[2]

**11. M/J 25/P24/Q6/a,b,c**

- 6 A battery of electromotive force (e.m.f.) 6.0 V and negligible internal resistance is connected in series with a variable resistor and a uniform resistance wire XY, as shown in Fig. 6.1.

**Fig. 6.1**

Wire XY has length 2.00 m and resistance  $8.0\,\Omega$ . The resistance  $R$  of the variable resistor is adjusted so that the potential difference across wire XY is 2.4 V.

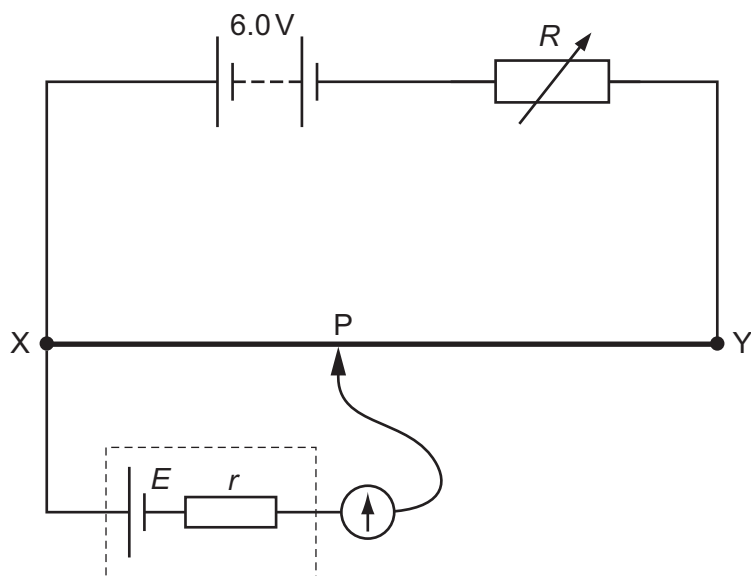
- (a) Determine  $R$ .

$$R = \quad \Omega \quad [2]$$

- (b) Explain why the potential difference  $V$  between any two points on wire XY is proportional to the distance  $L$  between those points.

[2]

- (c) A cell of e.m.f.  $E$  and internal resistance  $r$  is connected to the circuit, as shown in Fig. 6.2.

**Fig. 6.2**

Resistance  $R$  is unchanged.

The movable connection P is positioned on wire XY so that the galvanometer reading is zero. Distance XP is 1.24 m.

- (i) Calculate  $E$ .

$$E = \quad \text{V} \quad [2]$$

- (ii) The value of  $R$  is now decreased.

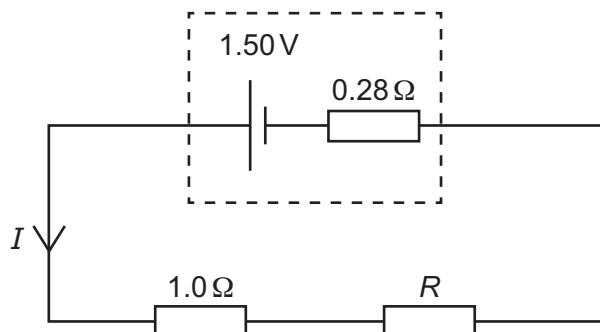
State and explain the change that must be made to the position of P on wire XY so that the galvanometer reads zero again.

[2]

**12. O/N 24/P21/Q7/a,b**

also Unit 9 (2 marks); this unit 6 marks

- 7 (a)** Fig. 7.1 shows two resistors connected in series with a cell of electromotive force (e.m.f.) 1.50 V and internal resistance  $0.28\ \Omega$ .

**Fig. 7.1**

One of the resistors has resistance  $1.0\ \Omega$ . The other resistor has resistance  $R$ . The terminal potential difference (p.d.) across the cell is 1.36 V.

- (i) Show that the current  $I$  in the circuit is 0.50 A.

[2]

- (ii) Calculate the combined resistance of the two resistors.

resistance =

Ω [2]

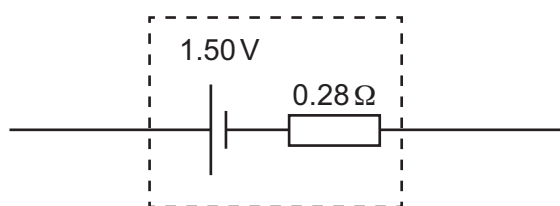
- (iii) Use your answer in (a)(ii) to determine resistance  $R$ .

$R =$

Ω [1]

- (b) The circuit in Fig. 7.1 is disconnected and the two resistors are reconnected to the cell, now in parallel with each other.

- (i) On Fig. 7.2, complete the circuit diagram to show this arrangement.

**Fig. 7.2**

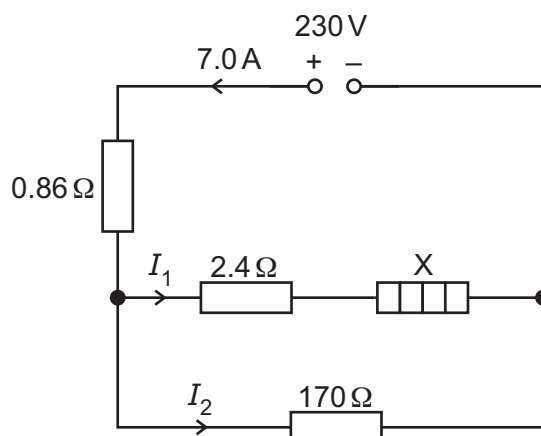
[1]

- (ii) Explain, without calculation, whether the terminal p.d. across the cell is now less than, equal to or greater than 1.36 V.

[2]

**13. O/N 24/P22/Q5/b(i,iii,iv,vii)**

- (b) A power supply, three resistors and a component X are connected in the circuit shown in Fig. 5.1.

**Fig. 5.1**

The power supply has an electromotive force (e.m.f.) of 230 V and negligible internal resistance. The current in the power supply is 7.0 A.

- (i) Identify component X.

[1]

- (iii) Determine the current  $I_1$ .

$$I_1 =$$

A [2]

- (iv) Calculate the p.d. across component X.

$$\text{p.d.} =$$

V [2]

- (vii) The resistor of resistance  $170\ \Omega$  is removed, leaving an open circuit in the lower branch of the circuit. There is no change to the resistance of component X.

State whether the current in the power supply increases, decreases or remains the same.

[1]

**14. O/N 24/P23/Q6/c**

- (b) A cylindrical metal wire of length 2.4 m and cross-sectional area  $8.0 \times 10^{-6} \text{ m}^2$  has a resistance of  $0.33 \Omega$ . There is a current in the wire of 4.7 A.

- (i) Determine the resistivity of the metal from which the wire is made.

resistivity =  $\Omega \text{ m}$  [2]

- (ii) Calculate the charge that passes through the wire in a time of 5.0 minutes.

charge =  $\text{C}$  [2]

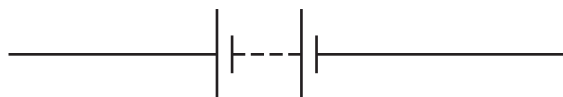
- (iii) The free electrons (charge carriers) in the wire have an average drift speed of  $0.16 \text{ mm s}^{-1}$ .

Determine the number density of charge carriers in the metal.

number density =  $\text{m}^{-3}$  [2]

- (c) The wire in (b) may be considered to be a fixed resistor. It is connected in series with a thermistor to a battery that has negligible internal resistance.

- (i) Use circuit symbols to complete Fig. 6.1 to show the circuit diagram of this arrangement.



**Fig. 6.1**

[1]

- (ii) Explain, without calculation, how the power dissipated in the wire changes as the temperature of the thermistor is increased.

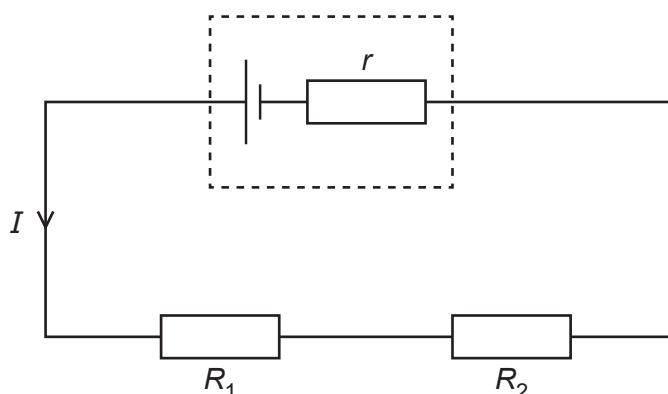
[2]

**15. M/J 24/P21/Q6/a,b,c,d**

- 6 (a) State Kirchhoff's first law.

[1]

- (b) A cell with internal resistance  $r$  is connected to two resistors of resistances  $R_1$  and  $R_2$  as shown in Fig. 6.1.



**Fig. 6.1**

The potential differences (p.d.s) across  $R_1$  and  $R_2$  are  $V_1$  and  $V_2$  respectively.  
 The terminal p.d. across the cell is  $V$ .  
 The current in the circuit is  $I$ .

Use Kirchhoff's laws to show that the total resistance  $R_T$  of the external circuit is given by

$$R_T = R_1 + R_2.$$

[2]

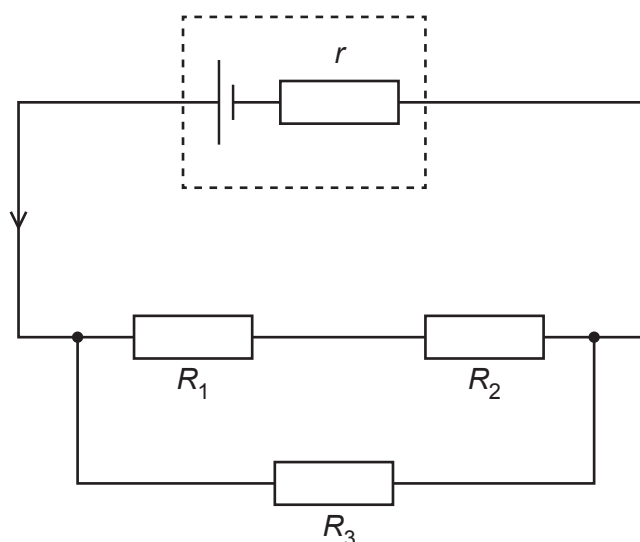
- (c) The electromotive force (e.m.f.) of the cell in Fig. 6.1 is 1.50 V.

The values of  $R_1$  and  $R_2$  are  $10\ \Omega$  and  $15\ \Omega$  respectively. The terminal p.d. of the cell is 1.35 V.

Calculate the internal resistance  $r$  of the cell.

$$r = \quad \quad \quad \Omega \quad [3]$$

- (d) A resistor of resistance  $R_3$  is added to the circuit in Fig. 6.1, so that the circuit is as shown in Fig. 6.2.



**Fig. 6.2**

State and explain the effect, if any, of this change on:

- (i) the current in the cell

[2]

- (ii) the terminal p.d. of the cell.

[2]

**16. M/J 24/P22/Q5/a,b**

- 5 (a) State Kirchhoff's second law.

[1]

- (b) A battery of electromotive force (e.m.f.) 9.0 V and negligible internal resistance is connected in series with a variable resistor X and a thermistor Y as shown in Fig. 5.1.

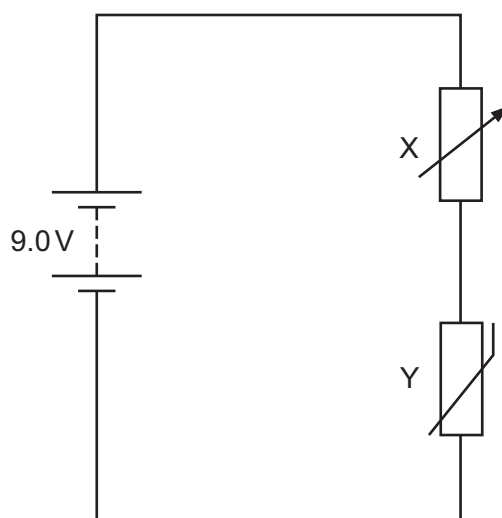


Fig. 5.1

Fig. 5.2 shows the relationship between temperature and resistance for the thermistor.

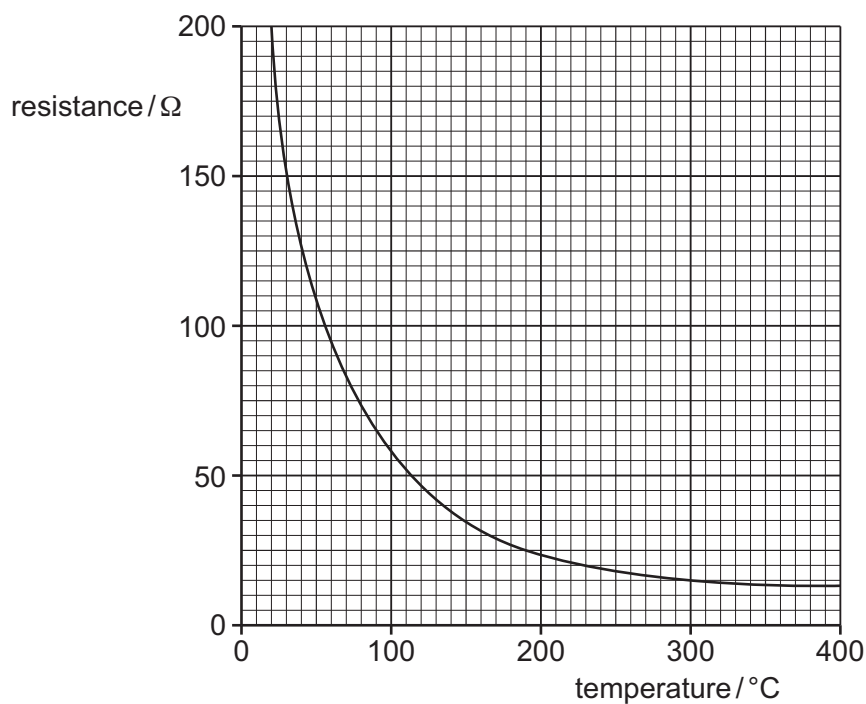


Fig. 5.2



- (i) The current in the circuit is  $1.1 \times 10^{-2} \text{ A}$ . The potential difference across Y is 4.0 V.

Calculate the resistance of X.

resistance = Ω [2]

- (ii) The temperature of Y is changed to 190 °C. The resistance of X remains unchanged.

Determine the new potential difference across Y.

potential difference = V [3]

- (iii) The resistance of X is increased. The temperature of Y remains at 190 °C.

By reference to the current in the circuit, state and explain the effect of this change, if any, on the potential difference **across Y**.

[3]

**17. M/J 24/P23/Q5/a,b,c**

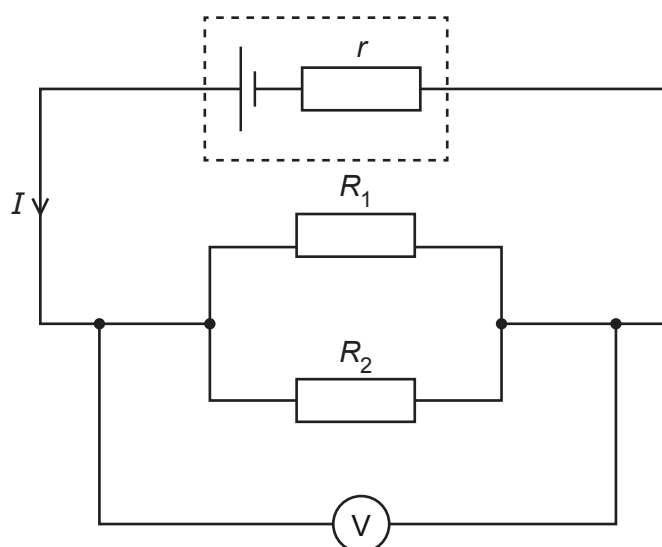
- 5 (a) (i)** State Kirchhoff's second law.

[1]

- (ii)** State the conservation law that gives rise to Kirchhoff's second law.

[1]

- (b)** A circuit contains a cell of internal resistance  $r$  and two resistors of resistances  $R_1$  and  $R_2$ , as shown in Fig. 5.1.

**Fig. 5.1**

The potential difference (p.d.) across the two resistors is  $V$ .

The current in the cell is  $I$ .

- (i)** Use Kirchhoff's laws to show that the total resistance  $R_T$  of the external circuit is given by

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2}.$$

[2]

- (ii)** The electromotive force (e.m.f.) of the cell is 1.50 V.

When the values of  $R_1$  and  $R_2$  are  $10\,\Omega$  and  $15\,\Omega$  respectively, the p.d. measured by the voltmeter is 1.38 V.

Calculate the internal resistance  $r$  of the cell.

$r =$

$\Omega$  [3]

- (c)** A third resistor is added in parallel with  $R_1$  and  $R_2$  in the circuit in Fig. 5.1.

State and explain the effect, if any, of this change on:

- (i)** the current in the cell

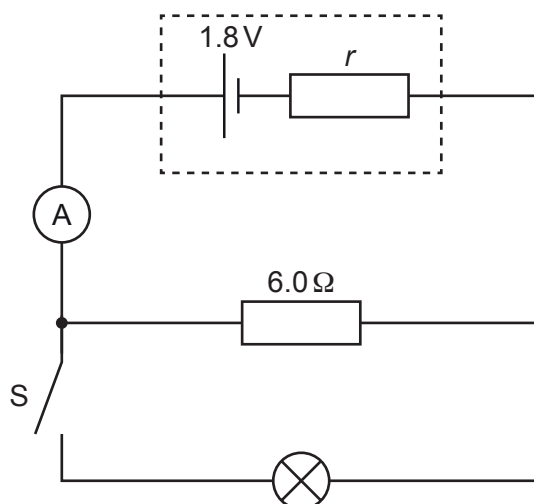
[2]

- (ii) the p.d. measured by the voltmeter.

[2]

**18. MAR 24/P22/Q7/b,c(i)**

- (b) A cell of electromotive force (e.m.f.)  $1.8\text{V}$  and internal resistance  $r$  is connected in parallel with a resistor of resistance  $6.0\Omega$  and a filament lamp, as shown in Fig. 7.1.



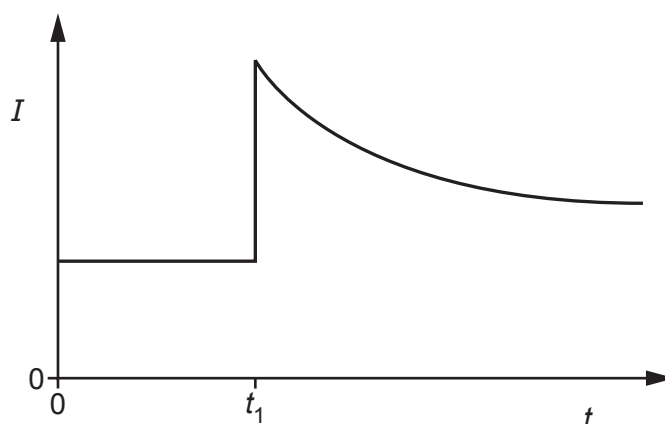
**Fig. 7.1**

The switch  $S$  is open. The ammeter reading is  $0.25\text{A}$ .

Determine the internal resistance  $r$  of the cell.

$$r = \quad \quad \quad \Omega \quad [3]$$

- (c) At time  $t_1$  switch  $S$  in Fig. 7.1 is closed. Fig. 7.2 shows the variation with time  $t$  of the ammeter reading  $I$ .



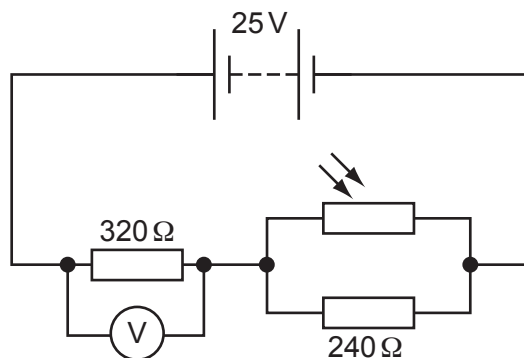
**Fig. 7.2**

- (i) State whether the e.m.f. of the cell after  $t_1$  is greater than, less than or the same as it was before  $t_1$ .

[1]

**19. O/N 23/P21/Q6/a(ii),b**

- 6** A battery is connected in a circuit with a light-dependent resistor (LDR), two fixed resistors and a voltmeter, as shown in Fig. 6.1.



**Fig. 6.1**

The battery has an electromotive force (e.m.f.) of 25 V and negligible internal resistance. The resistors have resistances of 320  $\Omega$  and 240  $\Omega$ .

- (a)** The voltmeter displays a reading of 16 V.

- (ii)** Calculate the resistance of the LDR.

resistance =

$\Omega$  [3]

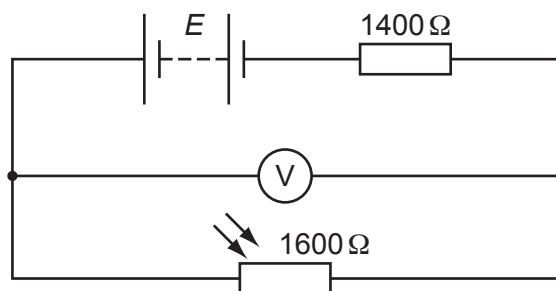
- (b)** The intensity of the light incident on the LDR increases.

State and explain what happens to the voltmeter reading.

[3]

**20. O/N 23/P22/Q6/b(iii),c(ii,iii)**

- (b) A battery of electromotive force (e.m.f.)  $E$  and negligible internal resistance is connected in series with a fixed resistor and a light-dependent resistor (LDR), as shown in Fig. 6.1.

**Fig. 6.1**

The resistance of the fixed resistor is  $1400\ \Omega$ . The intensity of the light illuminating the LDR causes it to have a resistance of  $1600\ \Omega$ . A voltmeter connected across the LDR reads  $6.4\text{ V}$ .

- (iii) Calculate the e.m.f.  $E$ .

$$E = \quad \quad \quad \text{V [2]}$$

- (c) The environmental conditions change causing a decrease in the resistance of the LDR in (b). The temperature of the environment does not change.

State whether there is a decrease, increase or no change to:

- (ii) the current in the battery

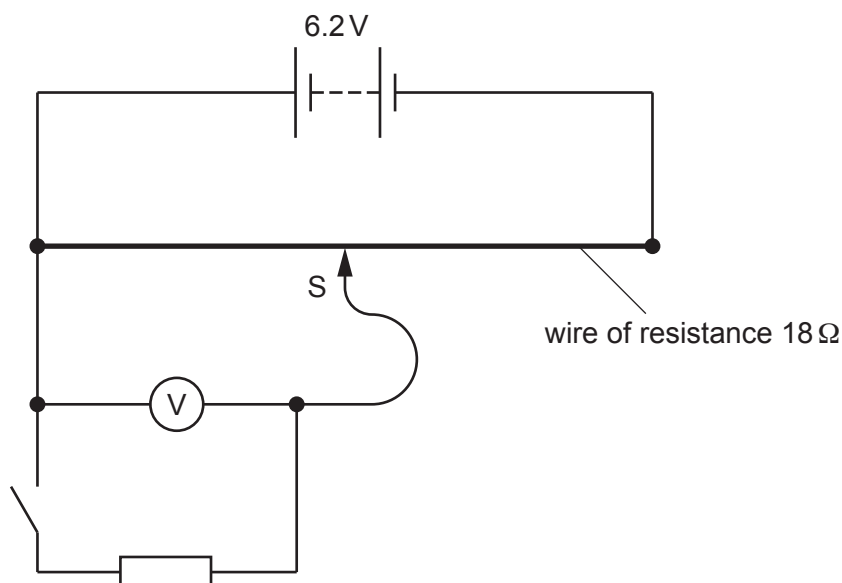
[1]

- (iii) the reading of the voltmeter.

[1]

**21. O/N 23/P23/Q7/b(ii,iii,iv)**

- (b) A battery of electromotive force (e.m.f.) 6.2V and negligible internal resistance is connected in a circuit to a uniform resistance wire, a voltmeter, a fixed resistor and a switch, as shown in Fig. 7.1.

**Fig. 7.1**

The resistance wire has resistance  $18\Omega$ , length  $0.94\text{ m}$  and cross-sectional area  $7.2 \times 10^{-8}\text{ m}^2$ . The slider S is positioned half-way along the length of the wire.

- (ii) The switch is open.  
State the reading on the voltmeter.

voltmeter reading = \_\_\_\_\_ V [1]

- (iii) The switch is now closed.

State whether there is an increase, decrease or no change to:

- the current in the battery
- the voltmeter reading.

[2]

- (iv) The switch remains closed. The slider S is moved along the resistance wire so that the voltmeter reading is  $3.1\text{ V}$ .

On Fig. 7.1, draw a cross (×) on the resistance wire to show a possible new position of the slider. [1]

**22. MAR 23/P22/Q6**

booklet (scan + OCR)

- (a) Define the potential difference across a component.

..... [1]

- (b) The variation with potential difference  $V$  of the current  $I$  in a semiconductor diode is shown in Fig. 6.1.

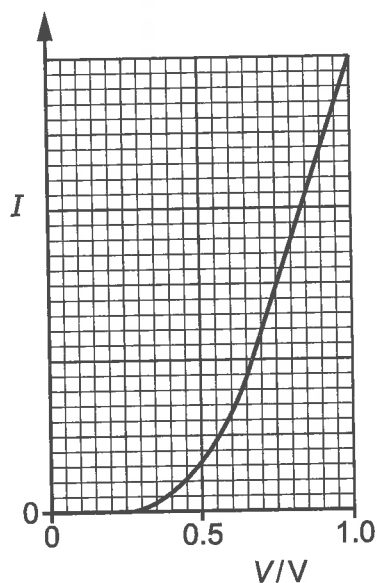


Fig. 6.1

Use Fig. 6.1 to describe qualitatively:

- (i) the resistance of the diode in the range  $V = 0$  to  $V = 0.25\text{ V}$

..... [1]

- (ii) the variation, if any, in the resistance of the diode as  $V$  changes from  $V = 0.75\text{ V}$  to  $V = 1.0\text{ V}$ .

..... [1]

- (c) A battery of electromotive force (e.m.f.) 12 V and negligible internal resistance is connected to a uniform resistance wire XY, a fixed resistor and a variable resistor, as shown in Fig. 6.2.

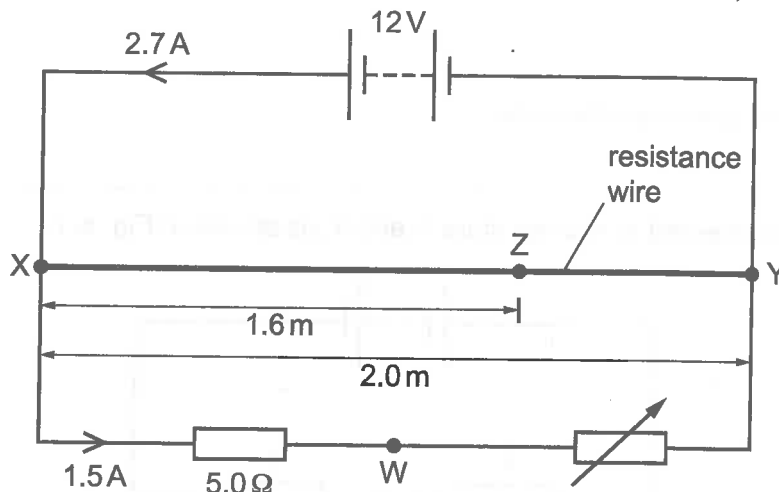


Fig. 6.2 (not to scale)

The fixed resistor has a resistance of  $5.0\Omega$ . The current in the battery is 2.7A and the current in the fixed resistor is 1.5A.

- (i) Calculate the current in the resistance wire.

current = .....A [1]

- (ii) Determine the resistance of the variable resistor.

resistance = .....  $\Omega$  [2]

- (iii) Wire XY has a length of 2.0m. Point Z on the wire is a distance of 1.6m from point X. The fixed resistor is connected to the variable resistor at point W. Determine the potential difference between points W and Z.

potential difference = ..... V [3]

- (iv) The resistance of the variable resistor is now increased.

By considering the currents in every part of the circuit, state and explain whether the total power produced by the battery decreases, increases or stays the same.

.....

.....

..... [3]

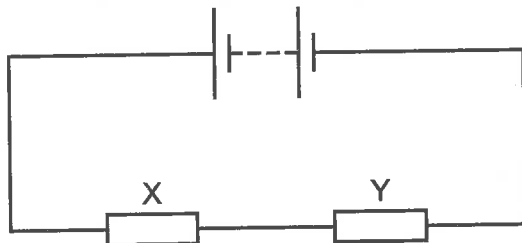
**23. O/N 22/P22/Q6**

booklet (scan + OCR)

- (a) Define electric potential difference.

[1]

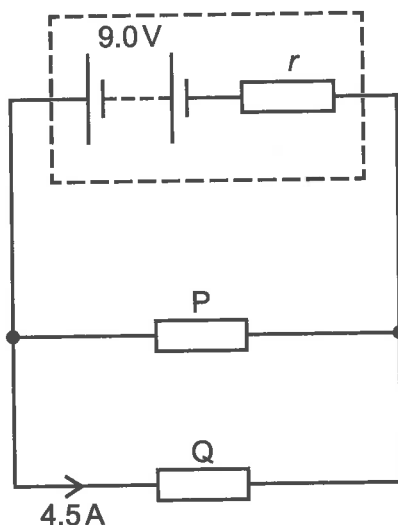
- (b) A battery is connected to two resistors X and Y, as shown in Fig. 6.1.

**Fig. 6.1**

The resistance of resistor X is greater than the resistance of resistor Y.  
State and explain which resistor dissipates more power.

[3]

- (c) A battery of electromotive force (e.m.f.) 9.0 V and internal resistance  $r$  is connected to two resistors P and Q, as shown in Fig. 6.2.

**Fig. 6.2**

A total charge of 650 C moves through resistor P in a time interval of 540 s. During this time resistor P dissipates 4800 J of energy. The current in resistor Q is 4.5 A. Assume that the e.m.f. of the battery remains constant.

Calculate:

(i) the current in resistor P

current = ..... A [2]

(ii) the potential difference across resistor P

potential difference = ..... V [2]

(iii) the internal resistance  $r$  of the battery.

$r$  = .....  $\Omega$  [2]

#### 24. O/N 22/P23/Q5

booklet (scan + OCR)

(a) State Kirchhoff's second law.

.....

..... [2]

(b) Three identical cells, each of electromotive force (e.m.f.) 1.5 V and internal resistance 590 m $\Omega$ , are connected in parallel across a conductor, as shown in Fig. 5.1.

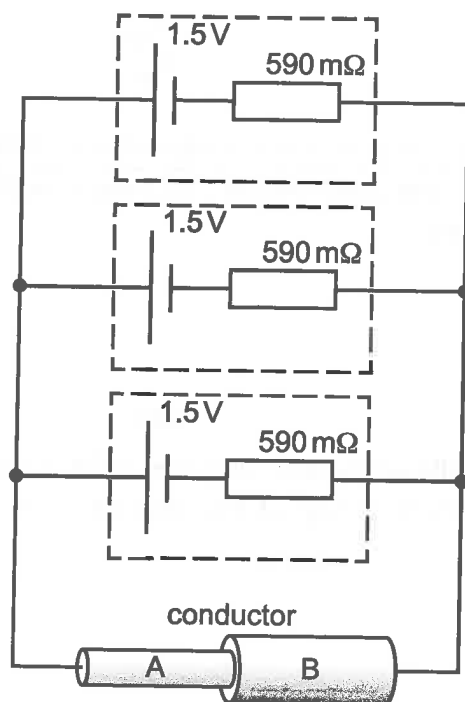


Fig. 5.1

The conductor is composed of two cylindrical sections A and B.  
The total resistance of the circuit is  $2.2\ \Omega$ .

- (i) Show that the resistance of the conductor is  $2.0\ \Omega$ .

[2]

- (ii) Calculate the current in the conductor.

current = ..... A [2]

- (c) The two cylindrical sections A and B of the conductor in Fig. 5.1 are made from the same material and have the same length.  
The diameter of section A is  $4.3\text{ mm}$  and the diameter of section B is  $7.6\text{ mm}$ .  
The resistance of section A is  $R_A$  and the resistance of section B is  $R_B$ .

- (i) Calculate the ratio  $\frac{R_A}{R_B}$ .

$\frac{R_A}{R_B} = \dots\dots\dots$  [3]

- (ii) Calculate the ratio  

$$\frac{\text{average drift speed of free electrons in section A}}{\text{average drift speed of free electrons in section B}}.$$
  
 Explain your reasoning.

ratio = ..... [2]

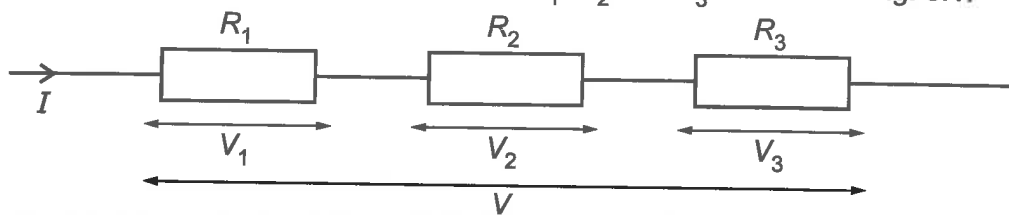
- (d) The circuit of Fig. 5.1 is altered by removing one of the cells.  
State and explain the effect, if any, of this change on the potential difference across the conductor.

.....  
 .....  
 ..... [3]

**25. M/J 22/P22/Q6**

booklet (scan + OCR)

- (a) A network of three resistors of resistances  $R_1$ ,  $R_2$  and  $R_3$  is shown in Fig. 6.1.

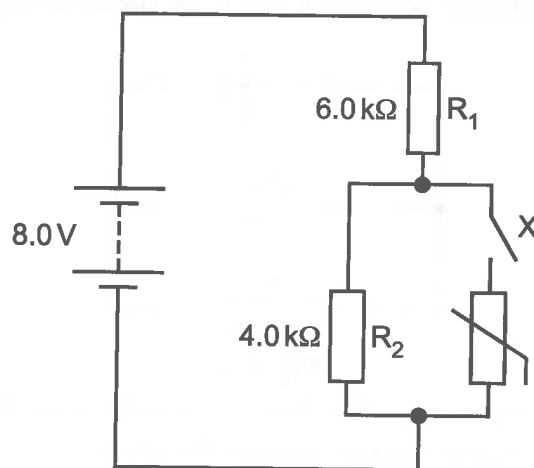
**Fig. 6.1**

The individual potential differences across the resistors are  $V_1$ ,  $V_2$  and  $V_3$ . The current in the combination of resistors is  $I$  and the total potential difference across the combination is  $V$ .

Show that the combined resistance  $R$  of the network is given by

$$R = R_1 + R_2 + R_3.$$

- (b) A battery of electromotive force (e.m.f.) 8.0V and negligible internal resistance is connected to a thermistor, a switch X and two fixed resistors, as shown in Fig. 6.2. [2]

**Fig. 6.2**

Resistor  $R_1$  has resistance  $6.0\text{ k}\Omega$  and resistor  $R_2$  has resistance  $4.0\text{ k}\Omega$ .

- (i) Switch X is open.

Calculate the potential difference across  $R_1$ .

potential difference = ..... V [2]

- (ii) Switch X is now closed. The resistance of the thermistor is  $12.0\text{ k}\Omega$ .  
Calculate the current in the battery.

current = ..... A [2]

- (c) The switch X in the circuit in (b) remains closed. The temperature of the thermistor decreases. By reference to the current in the battery, state and explain the effect, if any, of the decrease in temperature on the power produced by the battery.

.....  
 .....  
 ..... [3]

## 26. M/J 22/P21/Q6

booklet (scan + OCR)

- (a) State Kirchhoff's first law.

..... [1]

- (b) A battery is connected to two resistors X and Y, as shown in Fig. 6.1.

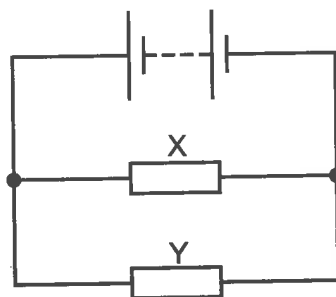


Fig. 6.1

The resistance of resistor X is greater than the resistance of resistor Y.

- (i) State and explain which resistor dissipates more power.

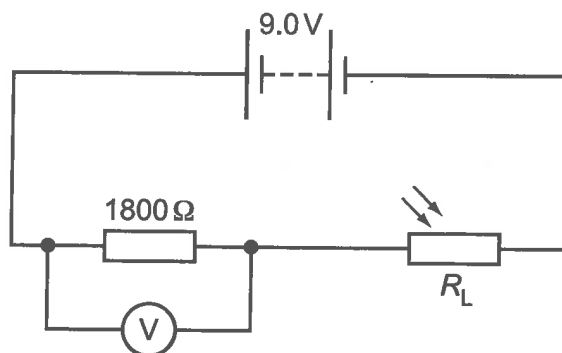
.....  
 .....  
 ..... [3]

- (ii) The two resistors are made of wires that have the same length. Both wires are made from metal of the same resistivity.

State and explain which resistor is made of wire with the larger cross-sectional area.

.....  
 ..... [2]

- (c) A battery of electromotive force (e.m.f.)  $9.0\text{ V}$  and negligible internal resistance is connected in series with a light-dependent resistor (LDR) and a fixed resistor of resistance  $1800\ \Omega$ , as shown in Fig. 6.2.



**Fig. 6.2**

A voltmeter is connected across the fixed resistor. The reading on the voltmeter is  $5.4\text{ V}$ .

- (i) Calculate the current in the circuit.

current = ..... A [1]

- (ii) Calculate the resistance  $R_L$  of the LDR.

$R_L = \dots\dots\dots\ \Omega$  [2]

- (iii) The intensity of the light illuminating the LDR increases.

By reference to the current in the circuit, state and explain the change, if any, to the voltmeter reading.

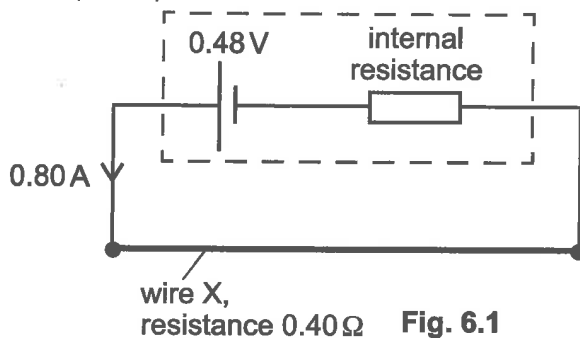
.....

..... [2]

## 27. O/N 21/P22/Q6

booklet (scan + OCR)

A cell of electromotive force (e.m.f.)  $0.48\text{ V}$  is connected to a metal wire X, as shown in Fig. 6.1.



**Fig. 6.1**

The cell has internal resistance. The current in the cell is  $0.80\text{ A}$ .

Wire X has length  $3.0\text{ m}$ , cross-sectional area  $1.3 \times 10^{-7}\text{ m}^2$  and resistance  $0.40\ \Omega$ .

- (a) Calculate the charge passing through the cell in a time of 7.5 minutes.

charge = ..... C [2]

- (b) Calculate the percentage efficiency with which the cell supplies power to wire X.

efficiency = ..... % [3]

- (c) There are  $3.2 \times 10^{22}$  free (conduction) electrons contained in the volume of wire X.

For wire X, calculate:

- (i) the number density  $n$  of the free electrons

$n = \dots\dots\dots \text{m}^{-3}$  [1]

- (ii) the average drift speed of the free electrons.

average drift speed = .....  $\text{m s}^{-1}$  [2]

- (d) A wire Y has the same cross-sectional area as wire X and is made of the same metal. Wire Y is longer than wire X.

Wire X in the circuit is now replaced by wire Y. Assume that wire Y has the same temperature as wire X.

State and explain whether the average drift speed of the free electrons in wire Y is greater than, the same as, or less than that in wire X.

.....  
 .....  
 ..... [3]

## 28. O/N 21/P21/Q5

booklet (scan + OCR)

- (a) State Kirchhoff's first law.

.....  
 ..... [2]

- (b) The circuit shown in Fig. 5.1 contains a battery of electromotive force (e.m.f.)  $E$  and negligible internal resistance connected to four resistors  $R_1$ ,  $R_2$ ,  $R_3$  and  $R_4$ , each of resistance  $R$ .

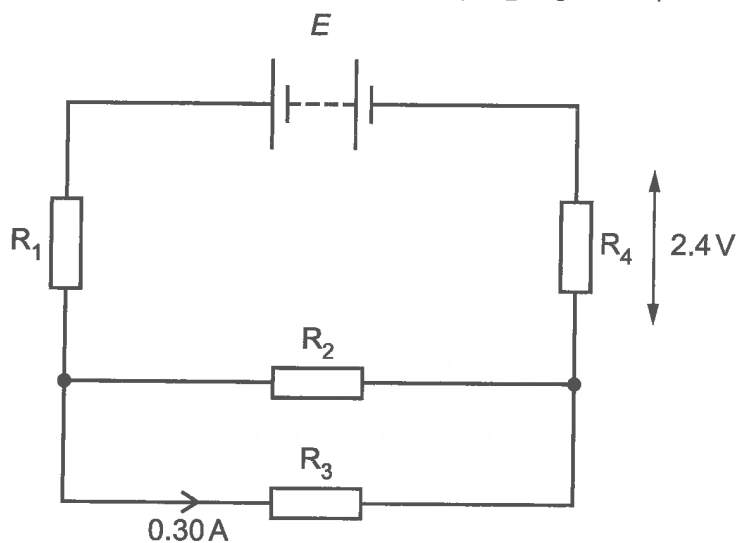


Fig. 5.1

The current in  $R_3$  is  $0.30\text{ A}$  and the potential difference (p.d.) across  $R_4$  is  $2.4\text{ V}$ .

- (i) Show that  $R$  is equal to  $4.0\ \Omega$ .

[2]

- (ii) Determine the e.m.f.  $E$  of the battery.

$$E = \dots\dots\dots \text{ V } [2]$$

- (c) The battery in (b) is replaced with another battery of the same e.m.f.  $E$  but with an internal resistance that is not negligible.

State and explain the change, if any, in the total power produced by the battery.

.....

..... [2]

- (d) The resistors in the circuit of Fig. 5.1 are made from nichrome wire of uniform radius  $240\ \mu\text{m}$ . The length of this wire needed to make each resistor is  $0.67\text{ m}$ .

Calculate the resistivity of nichrome.

$$\text{resistivity} = \dots\dots\dots \Omega\text{ m } [3]$$

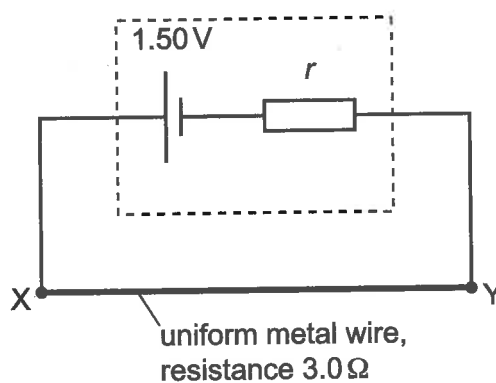
**29. O/N 21/P23/Q6**

booklet (scan + OCR)

- (a) A resistance wire of uniform cross-sectional area  $3.3 \times 10^{-7} \text{ m}^2$  and length 2.0 m is made of metal of resistivity  $5.0 \times 10^{-7} \Omega \text{ m}$ .  
Show that the resistance of the wire is  $3.0 \Omega$ .

[2]

- (b) The ends of the resistance wire in (a) are connected to the terminals X and Y in the circuit shown in Fig. 6.1.

**Fig. 6.1**

The cell has an electromotive force (e.m.f.) of 1.50 V and internal resistance  $r$ . The potential difference between X and Y is 1.20 V.

Calculate:

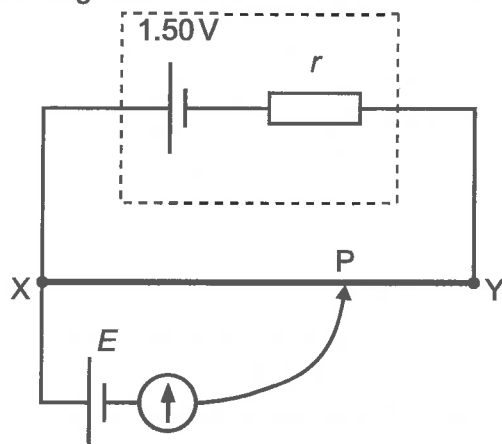
- (i) the current in the circuit

current = ..... A [1]

- (ii) the internal resistance  $r$ .

$r =$  .....  $\Omega$  [2]

- (c) A galvanometer and a cell of e.m.f.  $E$  with negligible internal resistance are connected to the circuit in (b), as shown in Fig. 6.2.



**Fig. 6.2**

The resistance wire between X and Y has a length of 2.0 m. The galvanometer has a reading of zero when the connection P is adjusted so that the length XP is 1.4 m.

Determine the e.m.f.  $E$  of the cell.

$$E = \dots\dots\dots \text{ V [2]}$$

- (d) The circuit in Fig. 6.2 is modified by replacing the original resistance wire with a second resistance wire. The second wire has the same length as the original wire and is made of the same metal.

The second wire has a smaller cross-sectional area than the original wire.

Connection P is adjusted on the second wire so that the galvanometer has a reading of zero. State and explain whether length XP for the second wire is shorter than, longer than or the same as length XP for the original wire when the galvanometer reading is zero.

.....  
 .....  
 ..... [3]

### 30. M/J 21/P21/Q5

booklet (scan + OCR)

- (a) State Kirchhoff's second law.

.....  
 ..... [2]

- (b) A battery has electromotive force (e.m.f.)  $4.0\text{ V}$  and internal resistance  $0.35\ \Omega$ . The battery is connected to a uniform resistance wire XY and a fixed resistor of resistance  $R$ , as shown in Fig. 5.1.

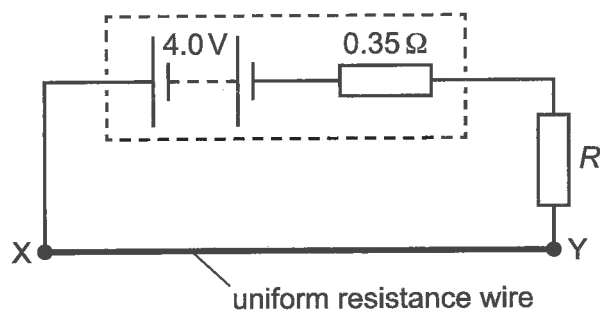


Fig. 5.1

Wire XY has resistance  $0.90\ \Omega$ . The potential difference across wire XY is  $1.8\text{ V}$ .

Calculate:

- (i) the current in wire XY

current = ..... A [1]

- (ii) the number of free electrons that pass a point in the battery in a time of  $45\text{ s}$

number = ..... [2]

- (iii) resistance  $R$ .

$R = \dots\dots\dots\ \Omega$  [2]

- (c) A cell of e.m.f.  $1.2\text{ V}$  is connected to the circuit in (b), as shown in Fig. 5.2.

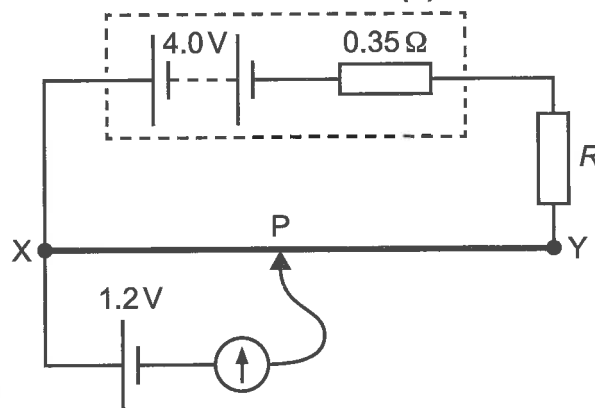


Fig. 5.2

The connection P is moved along the wire XY. The galvanometer reading is zero when distance XP is  $0.30\text{ m}$ .

- (i) Calculate the total length  $L$  of wire XY.

$$L = \dots\dots\dots \text{ m [2]}$$

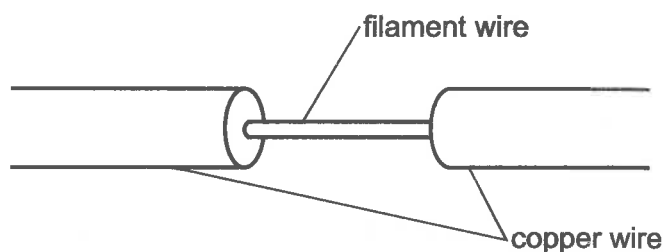
- (ii) The fixed resistor is replaced by a different fixed resistor of resistance greater than  $R$ . State and explain the change, if any, that must be made to the position of P on wire XY so that the galvanometer reading is zero.

.....  
 ..... [2]

### 31. MAR 21/P22/Q6/b(iv)

booklet (scan + OCR)

- (a) (i) The filament wire of the lamp is connected in series with the adjacent copper connecting wire of the circuit, as illustrated in Fig. 6.3.



**Fig. 6.3** (not to scale)

Some data for the filament wire and the adjacent copper connecting wire are given in Table 6.1.

**Table 6.1**

|                                  | filament wire | copper wire |
|----------------------------------|---------------|-------------|
| cross-sectional area             | $A$           | $360A$      |
| number density of free electrons | $n$           | $2.5n$      |

Calculate the ratio

$$\frac{\text{average drift speed of free electrons in filament wire}}{\text{average drift speed of free electrons in copper wire}}$$

$$\text{ratio} = \dots\dots\dots [2]$$

### 32. O/N 20/P22/Q6

booklet (scan + OCR)

- (a) A network of three resistors of resistances  $R_1$ ,  $R_2$  and  $R_3$  is shown in Fig. 6.1.

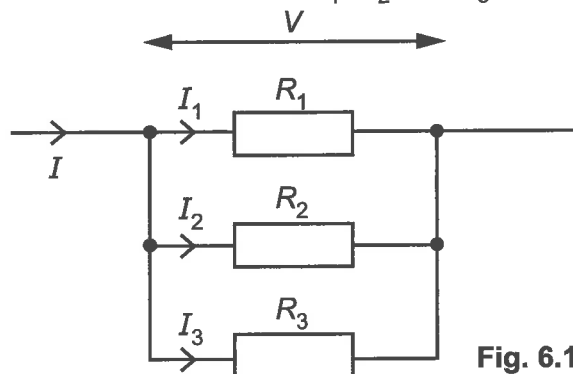


Fig. 6.1

The individual currents in the resistors are  $I_1$ ,  $I_2$  and  $I_3$ . The total current in the combination of resistors is  $I$  and the potential difference across the combination is  $V$ .

Show that the combined resistance  $R$  of the network is given by

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}.$$

[2]

- (b) A battery of electromotive force (e.m.f.) 8.0V and internal resistance  $r$  is connected to three resistors X, Y and Z, as shown in Fig. 6.2.

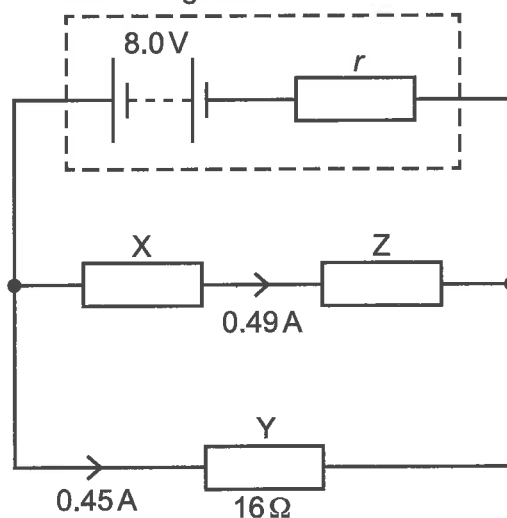


Fig. 6.2

Resistor Y has a resistance of  $16\Omega$ . The current in resistor X is 0.49A and the current in resistor Y is 0.45A.

Calculate:

- (i) the current in the battery

current = ..... A [1]

- (ii) the internal resistance  $r$  of the battery.

$$r = \dots\dots\dots \Omega \quad [2]$$

- (c) Resistors X and Y in Fig. 6.2 are made from wires of the same material and cross-sectional area. The average drift speed of the free electrons in X is  $2.1 \times 10^{-4} \text{ m s}^{-1}$ .

Calculate the average drift speed  $v$  of the free electrons in Y.

$$v = \dots\dots\dots \text{ m s}^{-1} \quad [2]$$

- (d) Resistor Z in Fig. 6.2 is replaced by a new resistor of smaller resistance.

State and explain the effect, if any, on the terminal potential difference of the battery.

.....  
 ..... [2]

### 33. O/N 20/P23/Q6

booklet (scan + OCR)

- (a) Define *electric potential difference* (*p.d.*).

..... [1]

- (b) A wire of cross-sectional area  $A$  is made from metal of resistivity  $\rho$ . The wire is extended. Assume that the volume  $V$  of the wire remains constant as it extends.

Show that the resistance  $R$  of the extending wire is inversely proportional to  $A^2$ .

[2]

- (c) A battery of electromotive force (e.m.f.)  $E$  and internal resistance  $r$  is connected to a variable resistor of resistance  $R$ , as shown in Fig. 6.1.

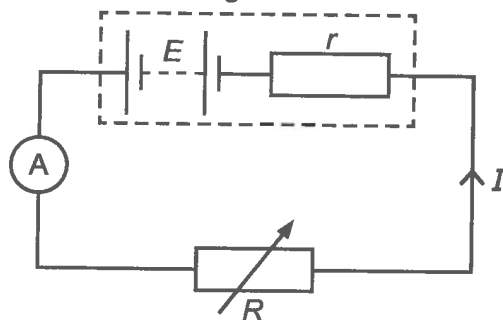


Fig. 6.1

The current in the circuit is  $I$ .

Use Kirchhoff's second law to show that

$$R = \left( \frac{E}{I} \right) - r.$$

[1]

- (d) An ammeter is used in the circuit in (c) to measure the current  $I$  as resistance  $R$  is varied.

Fig. 6.2 is a graph of  $R$  against  $\frac{1}{I}$ .

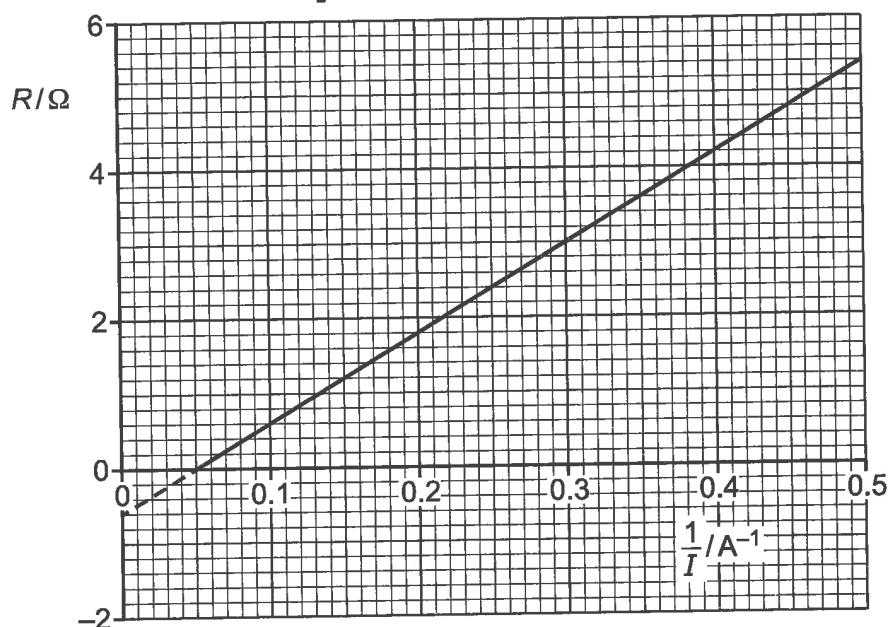


Fig. 6.2

- (i) Use Fig. 6.2 to determine the power dissipated in the variable resistor when there is a current of 2.0 A in the circuit.

power = ..... W [3]

- (ii) Use Fig. 6.2 and the equation in (c) to:

1. state the internal resistance  $r$  of the battery

$r = \dots \Omega$

2. determine the e.m.f.  $E$  of the battery.

$E = \dots \text{ V}$  [3]

### 34. M/J 20/P22/Q6

booklet (scan + OCR)

- (a) A battery of electromotive force (e.m.f.)  $7.8\text{V}$  and internal resistance  $r$  is connected to a filament lamp, as shown in Fig. 6.1.

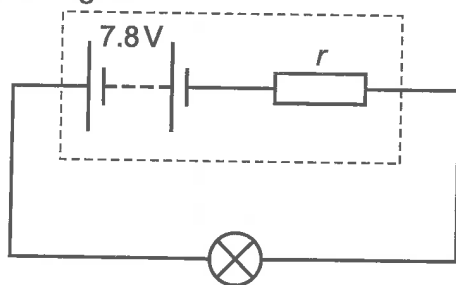


Fig. 6.1

A total charge of  $750\text{C}$  moves through the battery in a time interval of  $1500\text{s}$ . During this time the filament lamp dissipates  $5.7\text{kJ}$  of energy. The e.m.f. of the battery remains constant.

- (i) Explain, in terms of energy and without a calculation, why the potential difference across the lamp must be less than the e.m.f. of the battery.

..... [1]

- (ii) Calculate:

1. the current in the circuit

current = ..... A [2]

2. the potential difference across the lamp

potential difference = ..... V [2]

3. the internal resistance of the battery.

internal resistance = .....  $\Omega$  [2]

- (b) A student is provided with three resistors of resistances  $90\Omega$ ,  $45\Omega$  and  $20\Omega$ .

- (i) Sketch a circuit diagram showing how **two** of these three resistors may be connected together to give a combined resistance of  $30\Omega$  between the terminals shown. Label the values of the resistances on your diagram.



[1]

- (ii) A potential divider circuit is produced by connecting the three resistors to a battery of e.m.f.  $9.0\text{ V}$  and negligible internal resistance. The potential divider circuit provides an output potential difference  $V_{\text{OUT}}$  of  $3.6\text{ V}$ . The circuit diagram is shown in Fig. 6.2.

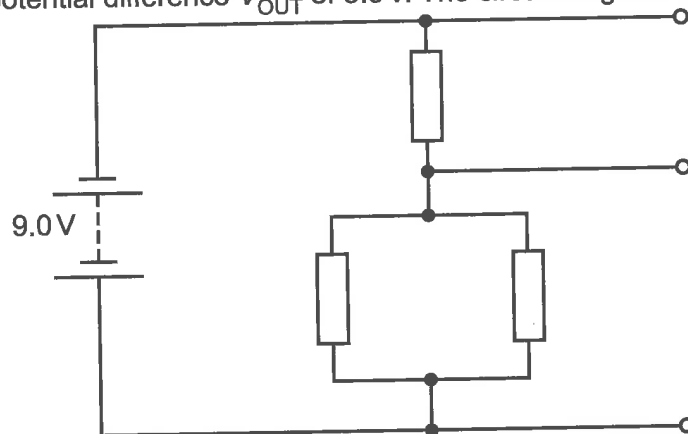


Fig. 6.2

On Fig. 6.2, label the resistances of all three resistors and the potential difference  $V_{\text{OUT}}$ . [2]

**35. M/J 20/P23/Q6**

booklet (scan + OCR)

The current  $I$  in a metal wire is given by the expression

$$I = Anve$$

where  $v$  is the average drift speed of the free electrons in the wire and  $e$  is the elementary charge.

(a) State what is meant by the symbols  $A$  and  $n$ .

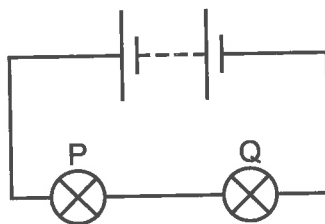
$A$ : .....

$n$ : ..... [2]

(b) Use the above expression to determine the SI base units of  $e$ .  
Show your working.

base units ..... [2]

(c) Two lamps P and Q are connected in series to a battery, as shown in Fig. 6.1.



**Fig. 6.1**

The radius of the filament wire of lamp P is twice the radius of the filament wire of lamp Q.  
The filament wires are made of metals with the same value of  $n$ .

Calculate the ratio

$$\frac{\text{average drift speed of free electrons in filament wire of P}}{\text{average drift speed of free electrons in filament wire of Q}}$$

ratio = ..... [2]

**36. MAR 20/P22/Q5**

booklet (scan + OCR)

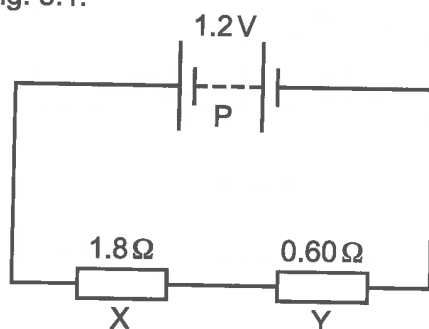
(a) Define the *ohm*.

..... [1]

(b) A wire has a resistance of  $1.8\ \Omega$ . The wire has a uniform cross-sectional area of  $0.38\ \text{mm}^2$  and is made of metal of resistivity  $9.6 \times 10^{-7}\ \Omega\text{m}$ .  
Calculate the length of the wire.

length = ..... m [3]

(c) A resistor X of resistance  $1.8\ \Omega$  is connected to a resistor Y of resistance  $0.60\ \Omega$  and a battery P, as shown in Fig. 5.1.



**Fig. 5.1**

The battery P has an electromotive force (e.m.f.) of  $1.2\text{ V}$  and negligible internal resistance.

(i) Explain, in terms of energy, why the potential difference (p.d.) across resistor X is less than the e.m.f. of the battery.

..... [1]

(ii) Calculate the potential difference across resistor X.

potential difference = ..... V [2]

- (d) Another battery Q of e.m.f. 1.2V and negligible internal resistance is now connected into the circuit of Fig. 5.1 to produce the new circuit shown in Fig. 5.2.

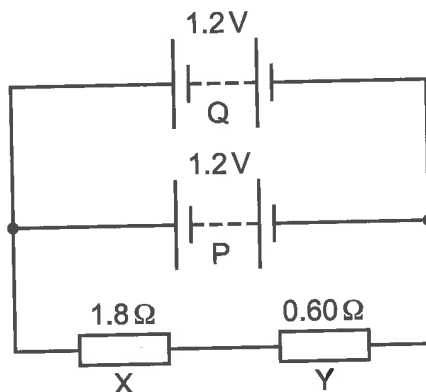


Fig. 5.2

State whether the addition of battery Q causes the current to decrease, increase or remain the same in:

- (i) resistor X ..... [1]
- (ii) battery P. .... [1]
- (e) The circuit shown in Fig. 5.2 is modified to produce the new circuit shown in Fig. 5.3.

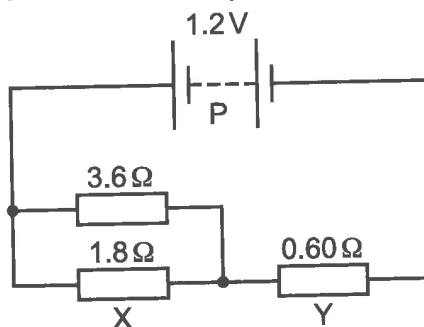


Fig. 5.3

Calculate:

- (i) the total resistance of the two resistors connected in parallel

resistance = .....  $\Omega$  [1]

- (ii) the current in resistor Y.

current = ..... A [2]

### 37. O/N 19/P22/Q6

booklet (scan + OCR)

(a) State Kirchhoff's first law.

..... [1]

(b) The variations with potential difference  $V$  of the current  $I$  for a resistor  $X$  and for a semiconductor diode are shown in Fig. 6.1.

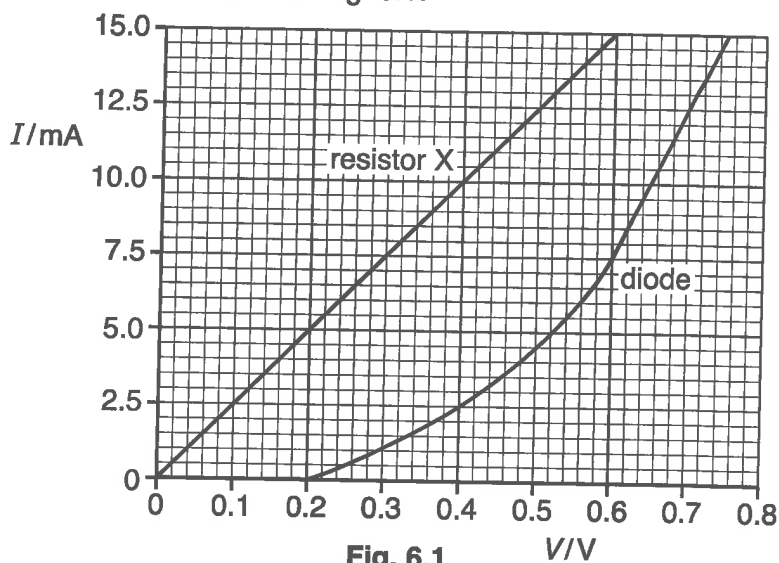


Fig. 6.1

(i) Determine the resistance of the diode for a potential difference  $V$  of 0.60 V.

resistance = .....  $\Omega$  [3]

(ii) Describe, qualitatively, the variation of the resistance of the diode as  $V$  increases from 0.60 V to 0.75 V.

..... [1]

(c) The diode and the resistor  $X$  in (b) are connected into the circuit shown in Fig. 6.2.

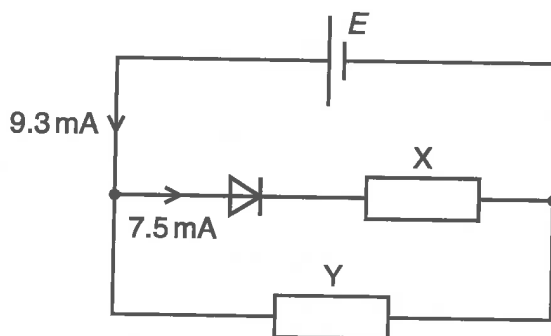


Fig. 6.2

The cell has electromotive force (e.m.f.)  $E$  and negligible internal resistance. Resistor Y is connected in parallel with resistor X and the diode. The current in the cell is  $9.3\text{ mA}$  and the current in the diode is  $7.5\text{ mA}$ .

(i) Use Fig. 6.1 to determine  $E$ .

$$E = \dots\dots\dots\text{V} \quad [1]$$

(ii) Determine the resistance of resistor Y.

$$\text{resistance} = \dots\dots\dots\Omega \quad [2]$$

(iii) Calculate the power dissipated in the diode.

$$\text{power} = \dots\dots\dots\text{W} \quad [2]$$

(iv) The cell is now replaced by a new cell of e.m.f.  $0.50\text{ V}$  and negligible internal resistance. Use Fig. 6.1 to determine the new current in the diode.

$$\text{current} = \dots\dots\dots\text{mA} \quad [1]$$

### 38. O/N 19/P23/Q6

booklet (scan + OCR)

A battery of electromotive force (e.m.f.)  $12\text{ V}$  and negligible internal resistance is connected to a network of two lamps and two resistors, as shown in Fig. 6.1.

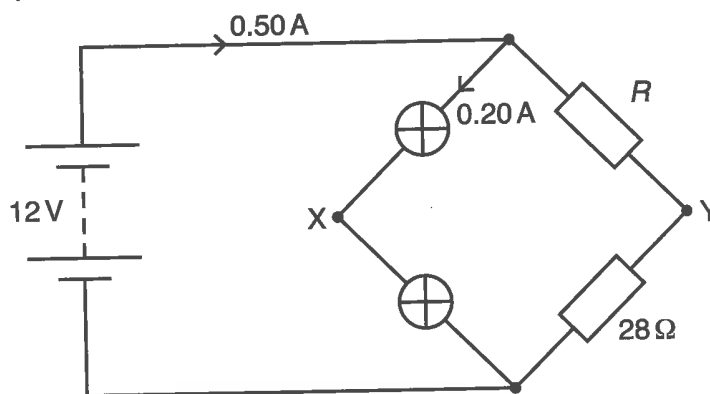


Fig. 6.1

The two lamps in the circuit have equal resistances. The two resistors have resistances  $R$  and  $28\Omega$ . The lamps are connected at junction X and the resistors are connected at junction Y. The current in the battery is  $0.50\text{ A}$  and the current in the lamps is  $0.20\text{ A}$ .

**(a)** Calculate:

**(i)** the resistance of each lamp

resistance = .....  $\Omega$  [2]

**(ii)** resistance  $R$ .

$R =$  .....  $\Omega$  [2]

**(b)** Determine the potential difference  $V_{XY}$  between points X and Y.

$V_{XY} =$  ..... V [3]

**(c)** Calculate the ratio

$$\frac{\text{total power dissipated by the lamps}}{\text{total power produced by the battery}}$$

ratio = ..... [2]

**(d)** The resistor of resistance  $R$  is now replaced by another resistor of lower resistance.

State and explain the effect, if any, of this change on the ratio in **(c)**.

.....  
 ..... [2]

### 39. M/J 19/P22/Q5

booklet (scan + OCR)

**(a)** State Kirchhoff's second law.

.....  
 ..... [2]

- (b) A battery of electromotive force (e.m.f.) 5.6 V and internal resistance  $r$  is connected to two external resistors, as shown in Fig. 5.1.

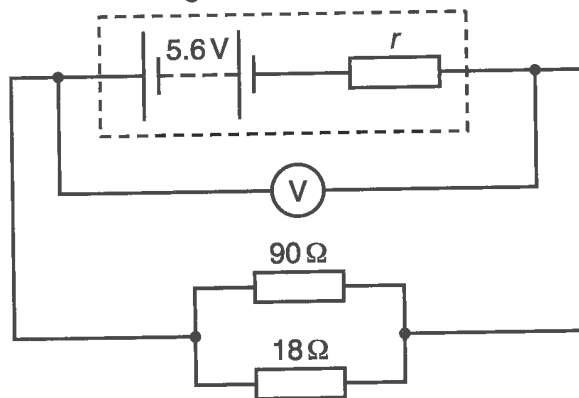


Fig. 5.1

The reading on the voltmeter is 4.8 V.

- (i) Calculate:

- the combined resistance of the two resistors connected in parallel

combined resistance = .....  $\Omega$  [2]

- the current in the battery.

current = ..... A [2]

- (ii) Show that the internal resistance  $r$  is  $2.5\ \Omega$ .

[2]

- (iii) Determine the ratio

$$\frac{\text{power dissipated by internal resistance } r}{\text{total power produced by battery}}.$$

ratio = ..... [3]

- (c) The battery in (b) is now connected to a battery of e.m.f.  $7.2\text{ V}$  and internal resistance  $3.5\ \Omega$ . The new circuit is shown in Fig. 5.2.

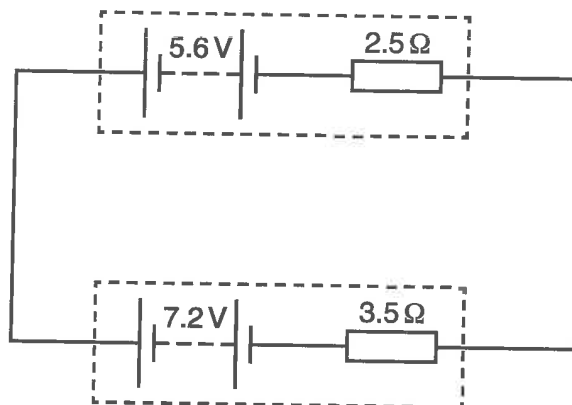


Fig. 5.2

Determine the current in the circuit.

current = ..... A [2]

#### 40. M/J 19/P23/Q6

booklet (scan + OCR)

- (a) Define the *ohm*.

..... [1]

- (b) A battery of electromotive force (e.m.f.)  $E$  and internal resistance  $1.5\ \Omega$  is connected to a network of resistors, as shown in Fig. 6.1.

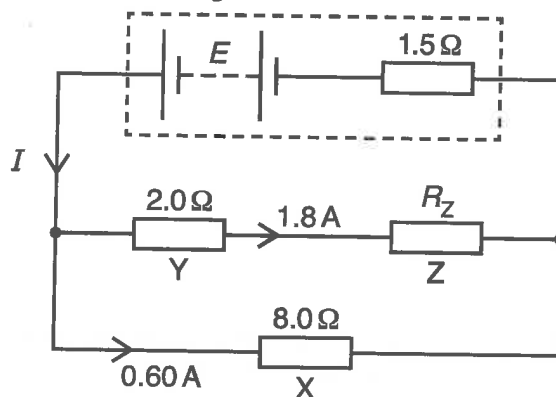


Fig. 6.1

Resistor X has a resistance of  $8.0\ \Omega$ . Resistor Y has a resistance of  $2.0\ \Omega$ . Resistor Z has a resistance of  $R_Z$ . The current in X is  $0.60\text{ A}$  and the current in Y is  $1.8\text{ A}$ .

- (i) Calculate:

1. the current  $I$  in the battery

$I = \dots\dots\dots\text{ A}$  [1]

2. resistance  $R_Z$

$$R_Z = \dots\dots\dots \Omega \text{ [2]}$$

3. e.m.f.  $E$ .

$$E = \dots\dots\dots \text{ V [2]}$$

(ii) Resistors X and Y are each made of wire. The two wires have the same length and are made of the same metal.

Determine the ratio:

1.  $\frac{\text{cross-sectional area of wire X}}{\text{cross-sectional area of wire Y}}$

$$\text{ratio} = \dots\dots\dots \text{ [2]}$$

2.  $\frac{\text{average drift speed of free electrons in X}}{\text{average drift speed of free electrons in Y}}$

$$\text{ratio} = \dots\dots\dots \text{ [2]}$$

#### 41. O/N 18/P22/Q6

booklet (scan + OCR)

(a) Define the *volt*.

$\dots\dots\dots$  [1]

(b) A battery of electromotive force (e.m.f.) 7.0 V and negligible internal resistance is connected in series with three components, as shown in Fig. 6.1.

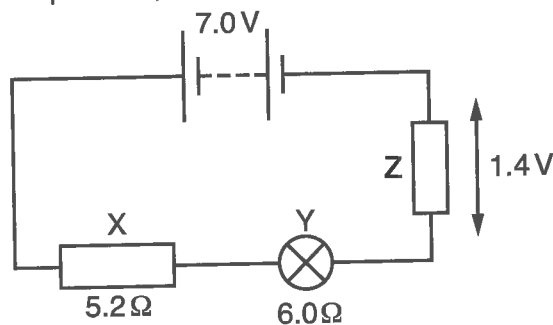


Fig. 6.1

Resistor X has a resistance of  $5.2\,\Omega$ . The resistance of the filament wire of lamp Y is  $6.0\,\Omega$ . The potential difference across resistor Z is  $1.4\,\text{V}$ .

(i) Calculate the current in the circuit.

current = ..... A [2]

(ii) Determine the resistance of resistor Z.

resistance = .....  $\Omega$  [1]

(iii) Calculate the percentage efficiency with which the battery supplies power to the lamp.

efficiency = ..... % [3]

(iv) The filament wire of the lamp is made of metal of resistivity  $3.7 \times 10^{-7}\,\Omega\,\text{m}$  at its operating temperature in the circuit.

Determine, for the filament wire, the value of  $\alpha$  where

$$\alpha = \frac{\text{cross-sectional area}}{\text{length}}.$$

$\alpha$  = ..... m [2]

#### 42. O/N 18/P21/Q6

booklet (scan + OCR)

(a) State Kirchhoff's second law.

.....  
 ..... [2]

- (b) An electric heater containing two heating wires X and Y is connected to a power supply of electromotive force (e.m.f.) 9.0 V and negligible internal resistance, as shown in Fig. 6.1.

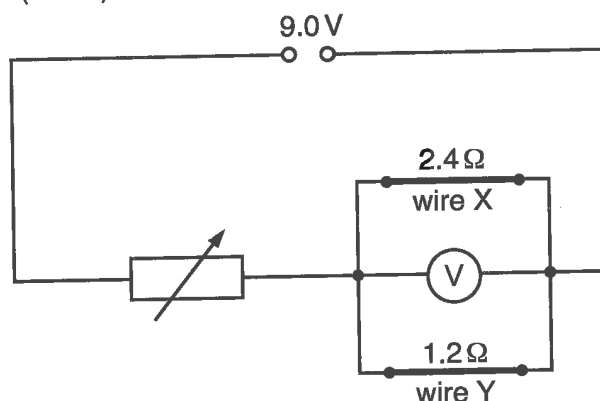


Fig. 6.1

Wire X has a resistance of  $2.4\ \Omega$  and wire Y has a resistance of  $1.2\ \Omega$ . A voltmeter is connected in parallel with the wires. A variable resistor is used to adjust the power dissipated in wires X and Y.

The variable resistor is adjusted so that the voltmeter reads 6.0 V.

- (i) Calculate the resistance of the variable resistor.

resistance = .....  $\Omega$  [3]

- (ii) Calculate the power dissipated in wire X.

power = ..... W [2]

- (iii) The cross-sectional area of wire X is three times the cross-sectional area of wire Y. Assume that the resistivity and the number density of free electrons for the metal of both wires are the same. Determine the ratio

1.  $\frac{\text{length of wire X}}{\text{length of wire Y}},$

ratio = ..... [2]

2.  $\frac{\text{average drift velocity of free electrons in wire X}}{\text{average drift velocity of free electrons in wire Y}}.$

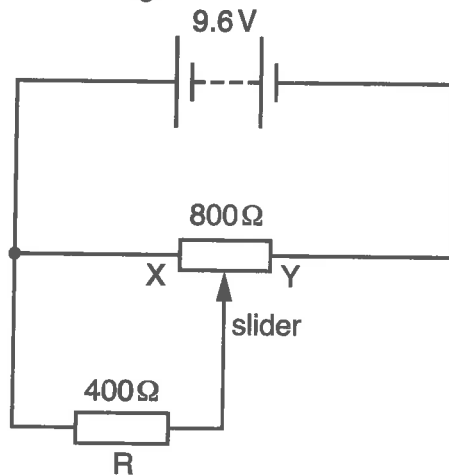
ratio = ..... [2]

**43. O/N 18/P23/Q7**

booklet (scan + OCR)

**(a)** State Kirchhoff's first law.

.....[1]

**(b)** A potentiometer is connected to a battery of electromotive force (e.m.f.) 9.6 V and negligible internal resistance, as shown in Fig. 7.1.**Fig. 7.1**

The maximum resistance of the potentiometer is  $800\ \Omega$ . A resistor R of resistance  $400\ \Omega$  is connected between the slider and end X of the potentiometer.

**(i)** State the potential difference across resistor R when the slider is positioned

1. at end X of the potentiometer,

potential difference = ..... V

2. at end Y of the potentiometer.

potential difference = ..... V  
[2]**(ii)** Calculate the potential difference across resistor R when the slider is positioned half-way between X and Y.

potential difference = ..... V [3]

**44. M/J 18/P22/Q6**

booklet (scan + OCR)

(a) (i) State Kirchhoff's first law.

..... [1]

(ii) Kirchhoff's first law is linked to the conservation of a certain quantity. State this quantity.

..... [1]

(b) A battery of electromotive force (e.m.f.)  $8.0\text{ V}$  and internal resistance  $2.0\Omega$  is connected to a resistor  $X$  and a wire  $Y$ , as shown in Fig. 6.1.

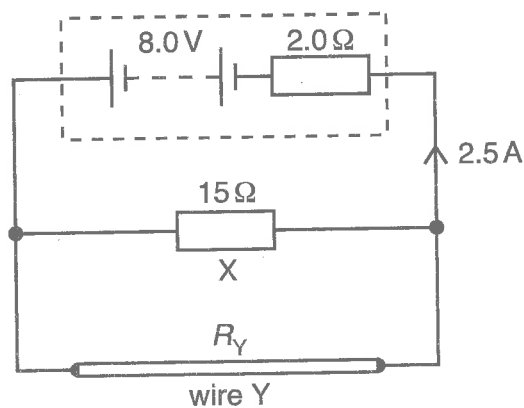


Fig. 6.1

The resistance of  $X$  is  $15\Omega$ . The resistance of  $Y$  is  $R_Y$ . The current in the battery is  $2.5\text{ A}$ .

(i) Calculate

- the thermal energy dissipated in the battery in a time of  $5.0$  minutes,

energy = ..... J [2]

- the terminal potential difference of the battery.

terminal potential difference = ..... V [1]

(ii) Determine the resistance  $R_Y$ .

$R_Y = \dots\dots\dots \Omega$  [3]

(iii) A new wire  $Z$  has the same length but less resistance than wire  $Y$ .

- State two possible differences between wire  $Z$  and wire  $Y$  that would separately cause wire  $Z$  to have less resistance than wire  $Y$ .

first difference: .....

second difference: ..... [2]

2. Wire Y is replaced in the circuit by wire Z. By considering the current in the battery, state and explain the effect of changing the wires on the total power produced by the battery.

..... [2]

**45. M/J 18/P23/Q6**

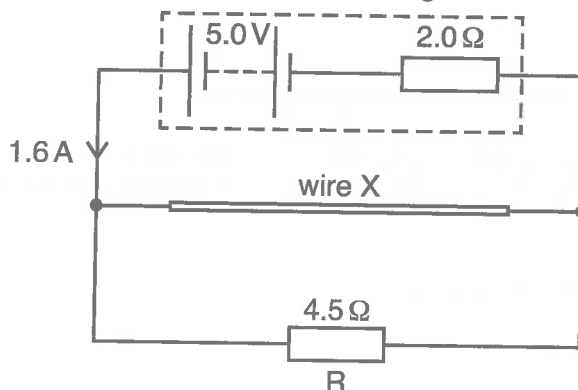
booklet (scan + OCR)

A wire X has a constant resistance per unit length of  $3.0 \Omega \text{ m}^{-1}$  and a diameter of  $0.48 \text{ mm}$ .

(a) Calculate the resistivity of the metal of wire X.

resistivity = .....  $\Omega \text{ m}$  [3]

(b) The wire X is connected into the circuit shown in Fig. 6.1.



**Fig. 6.1**

The battery has an electromotive force (e.m.f.) of  $5.0 \text{ V}$  and an internal resistance of  $2.0 \Omega$ . The wire X and a resistor R of resistance  $4.5 \Omega$  are connected in parallel. The current in the battery is  $1.6 \text{ A}$ .

(i) Calculate the potential difference across resistor R.

potential difference = .....  $\text{V}$  [1]

(ii) Determine, for wire X,  
1. its resistance,

resistance = .....  $\Omega$  [3]

2. its length.

length = .....  $\text{m}$  [1]

**46. MAR 18/P22/Q5**

booklet (scan + OCR)

(a) State Kirchhoff's second law.

.....  
 ..... [2]

(b) Two batteries, each of electromotive force (e.m.f.) 6.0 V and negligible internal resistance, are connected in series with three resistors, as shown in Fig. 5.1.

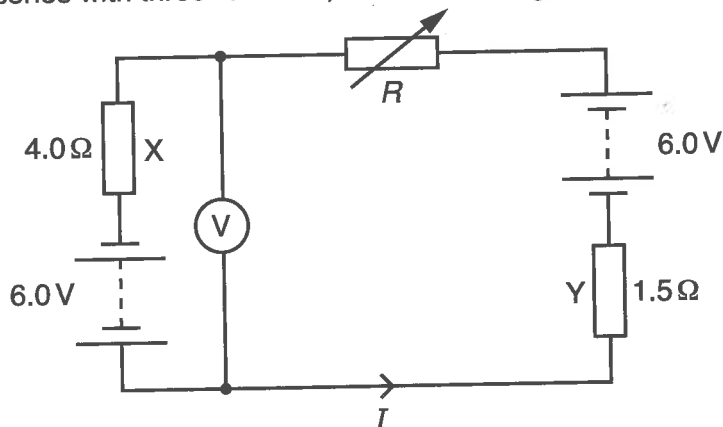


Fig. 5.1

Resistor X has resistance 4.0 Ω and resistor Y has resistance 1.5 Ω.

(i) The resistance  $R$  of the variable resistor is changed until the voltmeter in the circuit reads zero.

Calculate

1. the current  $I$  in the circuit,

$I = \dots\dots\dots$  A [1]

2. the resistance  $R$ .

$R = \dots\dots\dots$  Ω [2]

(ii) Resistors X and Y are wires made from the same material. The diameter of the wire of X is twice the diameter of the wire of Y.

Determine the ratio

$$\frac{\text{average drift speed of free electrons in X}}{\text{average drift speed of free electrons in Y}}$$

ratio =  $\dots\dots\dots$  [2]

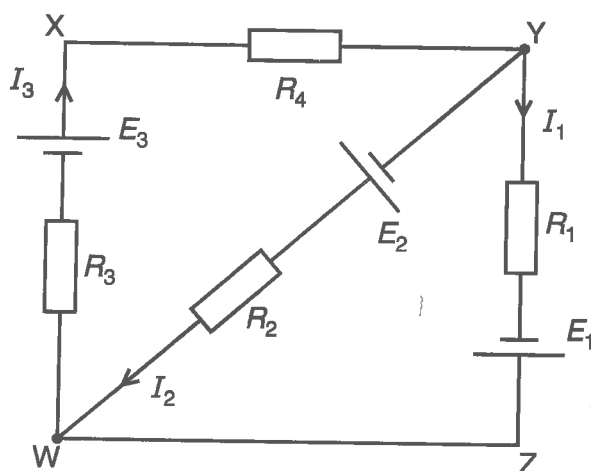
(iii) The resistance  $R$  of the variable resistor is now increased.  
 State and explain the effect of the increase in  $R$  on the power transformed by each of the batteries.

.....  
 .....  
 ..... [3]

**47. O/N 17/P21/Q5**

booklet (scan + OCR)

Three cells of electromotive forces (e.m.f.)  $E_1$ ,  $E_2$  and  $E_3$  are connected into a circuit, as shown in Fig. 5.1.

**Fig 5 1**

The circuit contains resistors of resistances  $R_1$ ,  $R_2$ ,  $R_3$  and  $R_4$ . The currents in the different parts of the circuit are  $I_1$ ,  $I_2$  and  $I_3$ . The cells have negligible internal resistance.

Use Kirchhoff's laws to state an equation relating

(a)  $I_1$ ,  $I_2$  and  $I_3$ ,

.....[1]

(b)  $E_1$ ,  $E_3$ ,  $R_1$ ,  $R_3$ ,  $R_4$ ,  $I_1$  and  $I_3$  in loop WXYZW,

.....[1]

(c)  $E_1$ ,  $E_2$ ,  $R_1$ ,  $R_2$ ,  $I_1$  and  $I_2$  in loop YZWY.

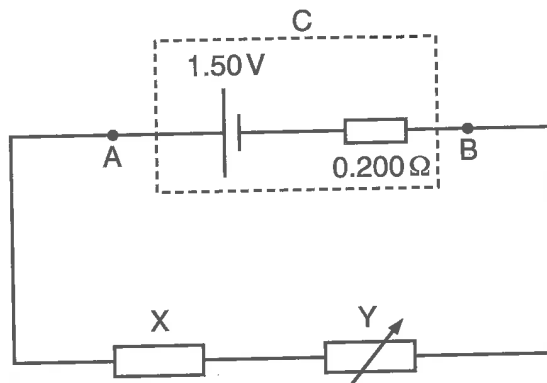
.....[1]

**48. M/J 17/P22/Q7**

booklet (scan + OCR)

**(a)** Define *electromotive force* (e.m.f.) of a cell.

..... [1]

**(b)** A cell C of e.m.f. 1.50 V and internal resistance  $0.200\ \Omega$  is connected in series with resistors X and Y, as shown in Fig. 7.1.**Fig. 7.1**

The resistance of X is constant and the resistance of Y can be varied.

**(i)** The resistance of Y is varied from 0 to  $8.00\ \Omega$ .

State and explain the variation in the potential difference (p.d.) between points A and B (terminal p.d. across C). Numerical values are not required.

.....

.....

..... [3]

**(ii)** The resistance of Y is set at  $6.00\ \Omega$ . The current in the circuit is  $0.180\text{ A}$ . Calculate

1. the resistance of X,

resistance = .....  $\Omega$  [2]

2. the p.d. between points A and B,

p.d. = ..... V [2]

3. the efficiency of the cell.

efficiency = ..... [2]

**49. O/N 16/P22/Q5**

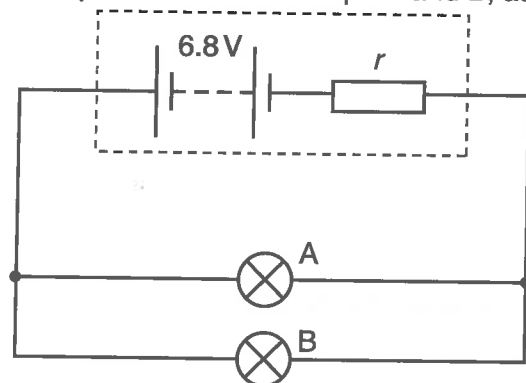
booklet (scan + OCR)

**(a)** State Kirchhoff's second law.

.....

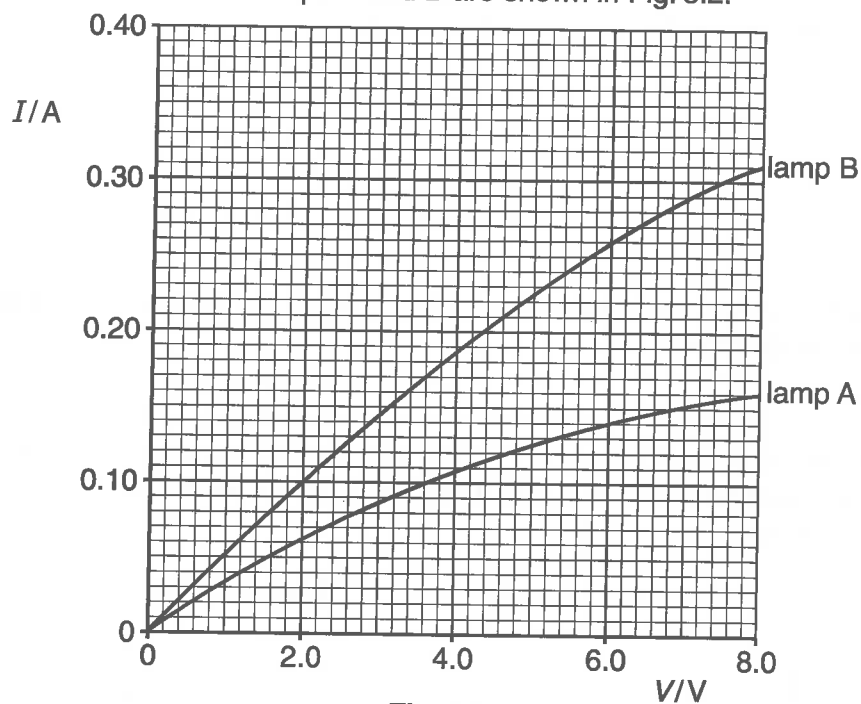
..... [2]

- (b) A battery is connected in parallel with two lamps A and B, as shown in Fig. 5.1.



**Fig. 5.1**

The battery has electromotive force (e.m.f.) 6.8V and internal resistance  $r$ . The  $I$ - $V$  characteristics of lamps A and B are shown in Fig. 5.2.



**Fig. 5.2**

The potential difference across the battery terminals is 6.0V.

- (i) Use Fig. 5.2 to show that the current in the battery is 0.40A.

[2]

- (ii) Calculate the internal resistance  $r$  of the battery.

$r = \dots\dots\dots \Omega$  [2]

(iii) Determine the ratio

$$\frac{\text{resistance of lamp A}}{\text{resistance of lamp B}}.$$

ratio = ..... [2]

(iv) Determine

1. the total power produced by the battery,

power = ..... W [2]

2. the efficiency of the battery in the circuit.

efficiency = ..... [2]

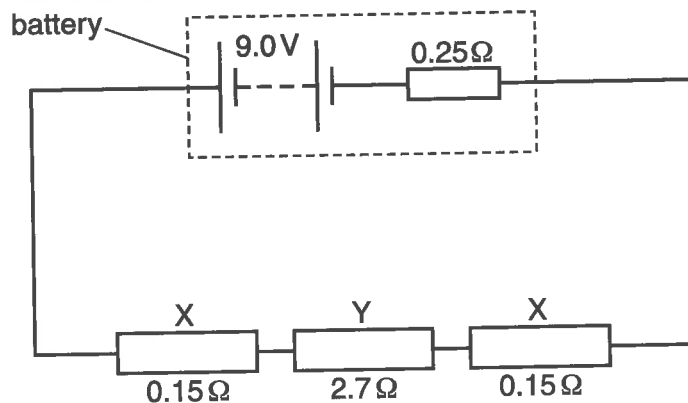
**50. M/J 16/P22/Q7**

booklet (scan + OCR)

- (a) Electric current is a flow of charge carriers. The charge on the carriers is quantised. Explain what is meant by *quantised*.

[1]

- (b) A battery of electromotive force (e.m.f.) 9.0V and internal resistance  $0.25\Omega$  is connected in series with two identical resistors X and a resistor Y, as shown in Fig. 7.1.

**Fig. 7.1**

The resistance of each resistor X is  $0.15\Omega$  and the resistance of resistor Y is  $2.7\Omega$ .

- (i) Show that the current in the circuit is 2.8A.

[3]

- (ii) Calculate the potential difference across the battery.

potential difference = ..... V [2]

- (c) Each resistor X connected in the circuit in (b) is made from a wire with a cross-sectional area of  $2.5\text{mm}^2$ . The number of free electrons per unit volume in the wire is  $8.5 \times 10^{29}\text{m}^{-3}$ .

- (i) Calculate the average drift speed of the electrons in X.

drift speed = .....  $\text{ms}^{-1}$  [2]

- (ii) The two resistors X are replaced by two resistors Z made of the same material and length but with half the diameter.

Describe and explain the difference between the average drift speed in Z and that in X.

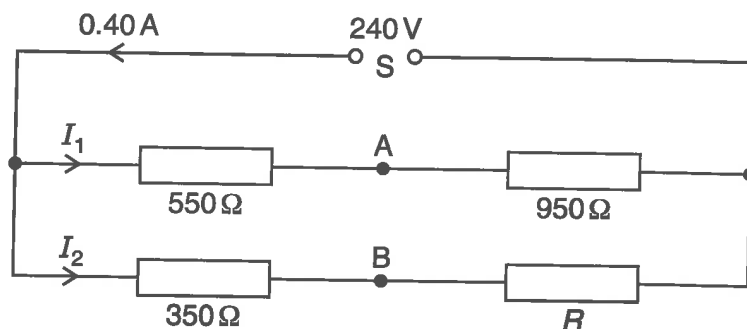
.....

..... [2]

**51. O/N 15/P22/Q5**

booklet (scan + OCR)

A 240 V power supply  $S$  with negligible internal resistance is connected to four resistors, as shown in Fig. 5.1.

**Fig. 5.1**

Two resistors of resistance  $550\ \Omega$  and  $950\ \Omega$  are connected in series across  $S$ . Two resistors of resistance  $350\ \Omega$  and  $R$  are also connected in series across  $S$ .

The current supplied by  $S$  is  $0.40\text{ A}$ .

Currents  $I_1$  and  $I_2$  in the circuit are shown in Fig. 5.1.

**(a)** Calculate

(i) current  $I_1$ ,

$$I_1 = \dots\dots\dots \text{ A [2]}$$

(ii) resistance  $R$ ,

$$R = \dots\dots\dots \Omega \text{ [2]}$$

(iii) the ratio

$$\frac{\text{power transformed in resistor of resistance } 350 \Omega}{\text{power transformed in resistor of resistance } 550 \Omega}.$$

$$\text{ratio} = \dots\dots\dots \text{ [2]}$$

(b) Two points are labelled A and B, as shown in Fig. 5.1.

(i) Calculate the potential difference  $V_{AB}$  between A and B.

$$V_{AB} = \dots\dots\dots \text{ V [2]}$$

(ii) The resistance  $R$  is increased.  
State and explain the effect on  $V_{AB}$ .

..... [1]

## 52. O/N 14/P22/Q5

booklet (scan + OCR)

A battery of electromotive force (e.m.f.) 12V and internal resistance  $r$  is connected in series to two resistors, each of constant resistance  $X$ , as shown in Fig. 5.1.

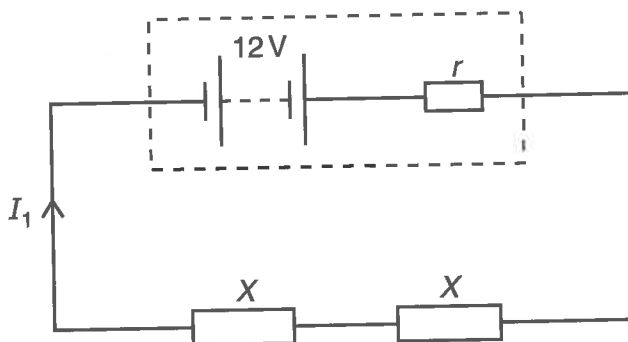


Fig. 5.1

The current  $I_1$  supplied by the battery is 1.2A.

The same battery is now connected to the same two resistors in parallel, as shown in Fig. 5.2.

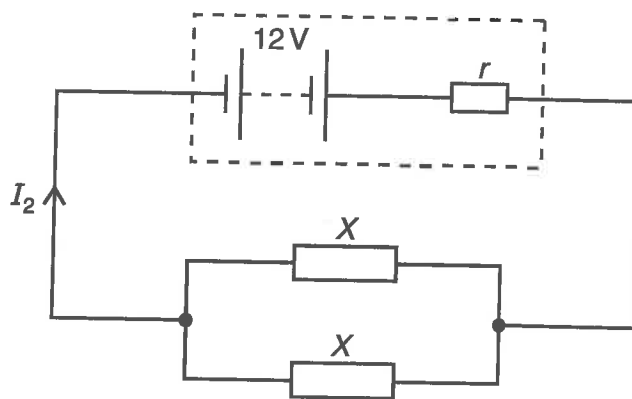


Fig. 5.2

The current  $I_2$  supplied by the battery is 3.0 A.

- (a) (i) Show that the combined resistance of the two resistors, each of resistance  $X$ , is four times greater in Fig. 5.1 than in Fig. 5.2.

- (ii) Explain why  $I_2$  is not four times greater than  $I_1$ .

[2]

.....

..... [2]

- (iii) Using Kirchhoff's second law, state equations, in terms of e.m.f., current,  $X$  and  $r$ , for

1. the circuit of Fig. 5.1,

.....

2. the circuit of Fig. 5.2.

.....

- (iv) Use the equations in (iii) to calculate the resistance  $X$ .

[2]

$$X = \text{.....} \Omega \quad [1]$$

- (b) Calculate the ratio

$$\frac{\text{power transformed in one resistor of resistance } X \text{ in Fig. 5.1}}{\text{power transformed in one resistor of resistance } X \text{ in Fig. 5.2}}$$

$$\text{ratio} = \text{.....} \quad [2]$$

- (c) The resistors in Fig. 5.1 and Fig. 5.2 are replaced by identical 12V filament lamps. Explain why the resistance of each lamp, when connected in series, is not the same as the resistance of each lamp when connected in parallel.

.....  
 ..... [2]

### 53. O/N 14/P21/Q7

booklet (scan + OCR)

- (a) A cell with internal resistance supplies a current. Explain why the terminal potential difference (p.d.) is less than the electromotive force (e.m.f.) of the cell.

..... [1]

- (b) A battery of e.m.f. 12V and internal resistance  $0.50\Omega$  is connected to a variable resistor X and a resistor Y of constant resistance, as shown in Fig. 7.1.

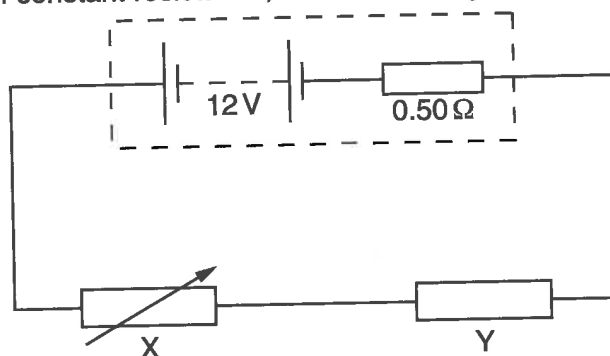


Fig. 7.1

The resistance  $R$  of X is increased from  $2.0\Omega$  to  $16\Omega$ . The variation with  $R$  of the current  $I$  in the circuit is shown in Fig. 7.2.

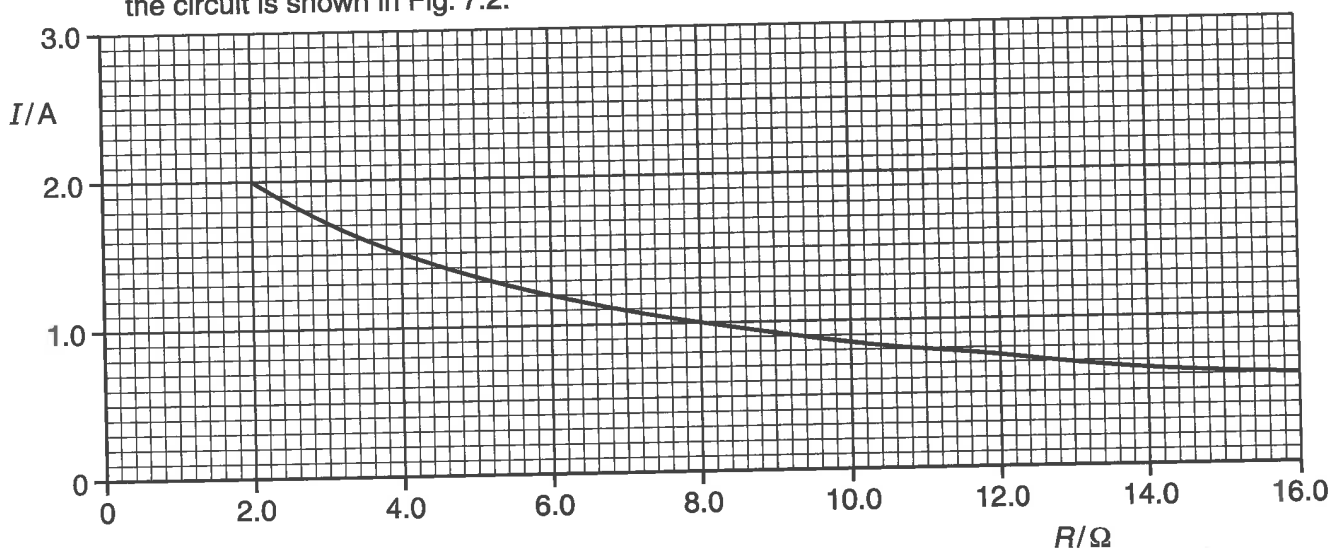


Fig. 7.2

Calculate, for  $I = 1.2\text{ A}$ ,  
 (i) the p.d. across X,

p.d. = ..... V [2]

(ii) the resistance of Y,

resistance = .....  $\Omega$  [3]

(iii) the power dissipated in the battery.

power = ..... W [2]

(c) Use Fig. 7.2 to explain the variation in the terminal p.d. of the battery as the resistance  $R$  of X is increased.

..... [1]

#### 54. O/N 14/P23/Q6

booklet (scan + OCR)

(a) A wire has length 100 cm and diameter 0.38 mm. The metal of the wire has resistivity  $4.5 \times 10^{-7} \Omega \text{ m}$ .

Show that the resistance of the wire is  $4.0 \Omega$ .

(b) The ends B and D of the wire in (a) are connected to a cell X, as shown in Fig. 6.1. [3]

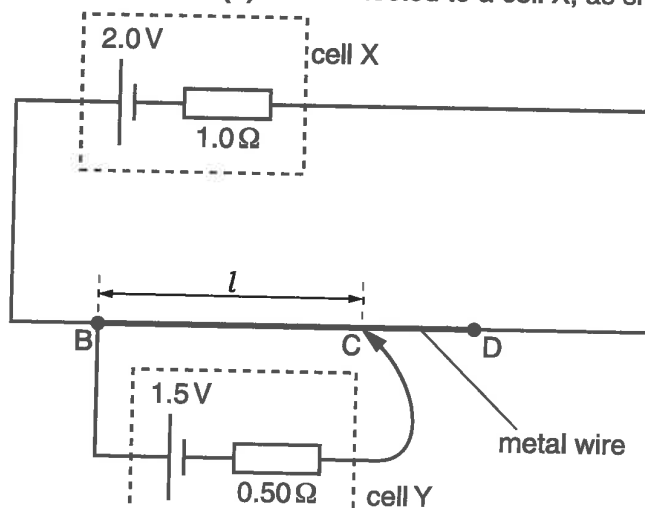


Fig. 6.1

The cell X has electromotive force (e.m.f.)  $2.0\text{V}$  and internal resistance  $1.0\Omega$ .

A cell Y of e.m.f.  $1.5\text{V}$  and internal resistance  $0.50\Omega$  is connected to the wire at points B and C, as shown in Fig. 6.1.

The point C is distance  $l$  from point B. The current in cell Y is zero.

Calculate

(i) the current in cell X,

current = ..... A [2]

(ii) the potential difference (p.d.) across the wire BD,

p.d. = ..... V [1]

(iii) the distance  $l$

$l$  = ..... cm [2]

(c) The connection at C is moved so that  $l$  is increased. Explain why the e.m.f. of cell Y is less than its terminal p.d.

.....  
.....[2]

## Answers Section

### 1. M/J 26/P22/Q5/a,b(i),c

|           |                                                                                                                         |             |                                                                                                                                                                                                                                                              |
|-----------|-------------------------------------------------------------------------------------------------------------------------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5(a)      | $I_T = I_1 + I_2 + I_3$<br>either $(V/R_T) = (V/R_1) + (V/R_2) + (V/R_3)$<br>or $(I_T/V) = (I_1/V) + (I_2/V) + (I_3/V)$ | <b>B1</b>   | Throughout allow $I$ and $R$ without subscript to mean $I_T$ and $R_T$ respectively, and allow $V_T$ for $V$                                                                                                                                                 |
|           | <u>and</u> $1/R_T = 1/R_1 + 1/R_2 + 1/R_3$                                                                              | <b>B1</b>   | If uses $V_1, V_2, V_3$ must also explicitly state that $(V =) V_1 = V_2 = V_3$ .<br>For the second mark, the associated working <u>and</u> final answer must both be shown<br>A fully correct answer in terms of two rather than three resistors credit 1/2 |
| 5(b)(i)   | $R_{(T)} = (1/2.8 + 1/2.8)^{-1}$<br>$= 1.4$<br>current in battery $= V/R_{(T)}$<br>$= 12/1.4$                           | <b>C1</b>   | Allow any correct expression for the parallel combination<br>e.g. $1/R_{(T)} = 1/2.8 + 1/2.8$ or $R_{(T)} = (2.8 \times 2.8)/(2.8 + 2.8)$                                                                                                                    |
|           | $= 8.6 \text{ A}$                                                                                                       | <b>A1</b>   |                                                                                                                                                                                                                                                              |
|           | OR<br>$I = 12/2.8$<br>$= 4.3$                                                                                           | <b>(C1)</b> |                                                                                                                                                                                                                                                              |
|           | current in battery $= 4.3 + 4.3$<br>$= 8.6 \text{ A}$                                                                   | <b>(A1)</b> | <u>Special case:</u><br>A final answer of 4.3 A without any supporting working credit 1/2                                                                                                                                                                    |
| 5(c)(i)   | light-dependent resistor                                                                                                | <b>B1</b>   | Allow LDR                                                                                                                                                                                                                                                    |
| 5(c)(ii)  | $I_{(X)} = 5 - 12/2.8$<br>$= 0.71$                                                                                      | <b>C1</b>   | Allow $I_{(X)} = 5 - (8.6/2)$ ( $= 0.70$ ) and allow ECF of current from <b>5(b)(i)</b>                                                                                                                                                                      |
|           | $R_{(\text{top branch})} = 12/0.71$<br>$= 16.8$                                                                         | <b>C1</b>   |                                                                                                                                                                                                                                                              |
|           | $R_{(X)} = 16.8 - 2.8$<br>$= 14 \Omega$                                                                                 | <b>A1</b>   |                                                                                                                                                                                                                                                              |
|           | OR<br>$I_{(X)} = 5 - 12/2.8$<br>$= 0.71$                                                                                | <b>(C1)</b> | Allow $I_{(X)} = 5 - (8.6/2)$ ( $= 0.70$ ) and allow ECF of current from <b>5(b)(i)</b>                                                                                                                                                                      |
|           | $V_{(\text{resistance wire})} = 0.71 \times 2.8$<br>$= 2.0$<br>$V_{(X)} = 12 - 2.0$<br>$= 10$                           | <b>(C1)</b> |                                                                                                                                                                                                                                                              |
|           | $R_{(X)} = 10/0.71$<br>$= 14 \Omega$                                                                                    | <b>(A1)</b> |                                                                                                                                                                                                                                                              |
| 5(c)(ii)  | OR<br>$R_{(T)} = 12/5$<br>$= 2.4$                                                                                       | <b>(C1)</b> |                                                                                                                                                                                                                                                              |
|           | $1/2.4 = 1/2.8 + 1/R_{(\text{top branch})}$<br>$R_{(\text{top branch})} = 16.8$                                         | <b>(C1)</b> | Allow $1/2.4 = 1/2.8 + 1/(2.8 + R_{(X)})$                                                                                                                                                                                                                    |
|           | $R_{(X)} = 16.8 - 2.8$<br>$= 14 \Omega$                                                                                 | <b>(A1)</b> |                                                                                                                                                                                                                                                              |
| 5(c)(iii) | (current in battery decreased, total resistance increased,) resistance (of X) increased                                 | <b>M1</b>   | Allow 'LDR' instead of X<br>Allow $V, I, R$ to mean potential difference, current and resistance respectively<br>Allow $\downarrow$ for decreases and $\uparrow$ for increases (BOD)<br>Ignore all comments about p.d. and emf                               |
|           | light intensity (incident on X) decreased                                                                               | <b>A1</b>   | Allow '(environment is) darker / dimmer / less light / less bright'<br>Allow 'temperature (of X/LDR) has decreased'                                                                                                                                          |

**2. M/J 26/P23/Q4/a,b(iii)**

|           |                                                                                                                                                                                                                       |           |                                                                                                                             |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------------------------------------------------------------------------------------------------------------------------|
| 4(a)      | correct circuit symbol used for cell with internal resistance                                                                                                                                                         | <b>B1</b> | Allow just a cell symbol, without the resistor. Ignore presence or absence of dotted box around it. not a battery symbol    |
|           | correct symbol for LDR connected to terminals of cell                                                                                                                                                                 | <b>B1</b> |                                                                                                                             |
|           | correct symbol for voltmeter connected in parallel with LDR                                                                                                                                                           | <b>B1</b> | If symbol for LDR incorrect, allow voltmeter connected in parallel with component drawn, provided only one component drawn. |
| 4(b)(iii) | $E = V + Ir$<br>$r = (4.90 - 4.78) \times 230 / 4.78$ or<br>$4.78 / 4.90 = 230 / (230 + r)$ or<br>$(4.90 - 4.78) / 4.90 = r / (230 + r)$ or<br>$(4.90 - 4.78) / 4.78 = r / 230$ or<br>$4.90 / (4.78 / 230) = r + 230$ | <b>C1</b> | Allow $E = I(R + r)$                                                                                                        |
|           | internal resistance = $5.8 \Omega$                                                                                                                                                                                    | <b>A1</b> |                                                                                                                             |

**3. M/J 26/P24/Q5/c,e**

|      |                                                                             |             |                                                        |
|------|-----------------------------------------------------------------------------|-------------|--------------------------------------------------------|
| 5(c) | (p.d. across component X =) 0.70 (from graph)                               | <b>C1</b>   | Allow 0.7                                              |
|      | (p.d. across $20 \Omega$ resistor = $0.005 \times 20$ ) = 0.10              | <b>C1</b>   | Allow 0.1<br>C1 marks are independent                  |
|      | (e.m.f. =) $0.70 + (0.005 \times 20) = 0.80$ (V)                            | <b>A1</b>   | Allow 0.8<br>Need 0.005 or $5 \times 10^{-3}$ not 5 mA |
|      | OR<br>(resistance of diode) = 140                                           | <b>(C1)</b> |                                                        |
|      | total resistance = $20 + 140$                                               | <b>(C1)</b> |                                                        |
|      | (e.m.f. =) $(0.7 / 5 \times 10^{-3} + 20) \times 5 \times 10^{-3} = 0.80$ V | <b>(A1)</b> | Allow 0.8                                              |
| 5(e) | (current in resistor =) $0.80 / 20 = 0.040$                                 | <b>C1</b>   | C1 marks are independent                               |
|      | (current in component X) = $0.040$ (A) / 40 (mA) (from graph)               | <b>C1</b>   | Allow 0.04 A                                           |
|      | current in ammeter = 0.080 A                                                | <b>A1</b>   | Allow 0.08 A                                           |
|      | or<br>resistance of diode is $0.8 / 40 \times 10^{-3} = 20 \Omega$          | <b>(C1)</b> |                                                        |
|      | total resistance in parallel is $10 \Omega$                                 | <b>(C1)</b> |                                                        |
|      | current = $0.8 / 10 = 0.080$ A                                              | <b>(A1)</b> | Allow 0.08 A                                           |

**4. MAR 26/P22/Q6/b,c(ii)**

|      |                                                           |           |
|------|-----------------------------------------------------------|-----------|
| 6(b) | a straight line from 0 A, 1.50 V with a negative gradient | <b>M1</b> |
|      | straight line passes through 1.60 A, 0.30 V               | <b>A1</b> |

|          |                                                                                           |             |
|----------|-------------------------------------------------------------------------------------------|-------------|
| 6(c)(ii) | terminal pd = $1.50 - (1.8 \times 0.75)$<br>= 0.15                                        | <b>C1</b>   |
|          | current in fixed resistor = $1.8 - (0.15 / 0.82)$<br>= 1.6                                | <b>C1</b>   |
|          | $R = 0.15 / 1.6$<br>= $0.093 \Omega$                                                      | <b>A1</b>   |
|          | OR<br>terminal pd = $1.50 - (1.8 \times 0.75)$<br>= 0.15                                  | <b>(C1)</b> |
|          | total external resistance = $0.15 / 1.8$<br>= 0.083<br><br>$1 / 0.083 = 1 / 0.82 + 1 / R$ | <b>(C1)</b> |
|          | $R = 0.093 \Omega$                                                                        | <b>(A1)</b> |

**5. O/N 25/P21/Q5/a,b(ii),c**

|          |                                                                                                                                                    |           |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 5(a)     | <u>sum of</u> current(s) into junction = <u>sum of</u> current(s) out junction<br><b>or</b><br>(algebraic) sum of current(s) at a junction is zero | <b>B1</b> |
| 5(b)(ii) | (resistance of T decreases so) total resistance (of circuit) decreases                                                                             | <b>B1</b> |
|          | current in cell increases (so p.d. across internal resistance increases)                                                                           | <b>B1</b> |
|          | terminal p.d. decreases (and resistance of R is constant) so current in R decreases                                                                | <b>B1</b> |
| 5(c)(i)  | p.d. across $r$ = $1.50 - (6.00 \times 0.200)$<br>(= 0.30 V)                                                                                       | <b>C1</b> |
|          | current in cell = $0.30 / 0.12$<br>= 2.5 A                                                                                                         | <b>A1</b> |
| 5(c)(ii) | current in thermistor = $2.5 - 0.200$<br>(= 2.3 A)                                                                                                 | <b>C1</b> |
|          | resistance of T = $(6.00 \times 0.200) / 2.3$<br>= $0.52 \Omega$                                                                                   | <b>A1</b> |

**6. O/N 25/P22/Q5/a,b(i,ii,iv)**

|         |                                                                                                                                              |           |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 5(a)    | <u>sum of</u> current(s) into junction = <u>sum of</u> current(s) out junction <b>or</b> (algebraic) sum of current(s) at a junction is zero | <b>B1</b> |
| 5(b)(i) | $(V =) E - Ir$                                                                                                                               | <b>C1</b> |
|         | $(V =) 5.0 - (38 \times 10^{-3} \times 4.7) = 4.8 \text{ (V)}$                                                                               | <b>A1</b> |

|          |                                                                                                              |             |
|----------|--------------------------------------------------------------------------------------------------------------|-------------|
| 5(b)(ii) | $R_{(Z)} = 120 \, \Omega$                                                                                    | <b>C1</b>   |
|          | $I_2 = 38 \times 10^{-3} - (4.8 / (100 + 120))$                                                              | <b>C1</b>   |
|          | $I_2 = 0.016 \, \text{A}$                                                                                    | <b>A1</b>   |
|          | <b>or</b>                                                                                                    |             |
|          | $R_{(Z)} = 120 \, \Omega$                                                                                    | <b>(C1)</b> |
|          | $R_{(\text{EXT})} = 4.8 / 38 \times 10^{-3}$<br>(= 126 $\Omega$ )                                            | <b>(C1)</b> |
|          | $1 / R_{(V)} = 1 / 126 - 1 / (120 + 100)$<br>( $R_{(V)} = 297 \, \Omega$ )                                   |             |
|          | $I_2 = 4.8 / 297$<br>$I_2 = 0.016 \, \text{A}$                                                               | <b>(A1)</b> |
| 5(b)(iv) | resistance of the LDR / Z increases (as light intensity decreases)                                           | <b>B1</b>   |
|          | <u>total</u> resistance (of the circuit) increases<br><b>or</b><br>current <u>in the battery</u> decreases   | <b>M1</b>   |
|          | (potential difference across the internal resistor decreases so) the terminal potential difference increases | <b>A1</b>   |

**7. O/N 25/P24/Q5/a(ii)**

|          |                                                                                                        |           |
|----------|--------------------------------------------------------------------------------------------------------|-----------|
| 5(a)(ii) | resistor connected to cell in a closed loop <b>and</b> correct circuit symbols used for all components | <b>B1</b> |
|          | ammeter connected in series with resistor <b>and</b> voltmeter connected in parallel with resistor     | <b>B1</b> |

**8. M/J 25/P21/Q6/b(i)**

|         |                                                                            |           |
|---------|----------------------------------------------------------------------------|-----------|
| 6(b)(i) | (Total) resistance decreases (and the potential difference stays the same) | <b>M1</b> |
|         | (so the reading on the ammeter) increases                                  | <b>A1</b> |

**9. M/J 25/P22/Q6/b(ii,iii,iv),c**

|           |                                                                                                                                                               |             |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| 6(b)(ii)  | $I = V / R$<br>$I_1 = 8.0 / (430 + 240)$                                                                                                                      | <b>C1</b>   |
|           | = 0.012 A                                                                                                                                                     | <b>A1</b>   |
| 6(b)(iii) | $I_2 = 0.030 - I_1$<br>= 0.030 - 0.012<br>= 0.018 A                                                                                                           | <b>A1</b>   |
| 6(b)(iv)  | $R = V / I_2$<br>= $(8.0 - (0.018 \times 210)) / 0.018$                                                                                                       | <b>C1</b>   |
|           | = 230 $\Omega$                                                                                                                                                | <b>A1</b>   |
|           | OR<br>resistance of top branch = $8.0 / 0.018$<br>$R = 8.0 / 0.018 - 210$                                                                                     | <b>(C1)</b> |
|           | = 230 $\Omega$                                                                                                                                                | <b>(A1)</b> |
|           | OR<br>total circuit resistance = $8.0 / 0.030 = 267$<br>$1 / 267 = 1 / (210 + R) + 1 / (430 + 240)$<br>$1 / 267 - 1 / 670 = 1 / (210 + R)$<br>$210 + R = 443$ | <b>(C1)</b> |
|           | $R = 230 \, \Omega$                                                                                                                                           | <b>(A1)</b> |

|      |                                                                                                                                                        |             |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| 6(c) | (When) the galvanometer reads 0 (A)                                                                                                                    | <b>M1</b>   |
|      | The ratio of the resistances in the top branch will equal the ratio of the resistances in the bottom branch (so the resistance of X can be determined) | <b>A1</b>   |
|      | OR<br>The ratio of the left pair of resistances will equal the ratio of the right pair of resistances                                                  | <b>(A1)</b> |

**10. M/J 25/P23/Q7/b**

|           |                                                                                                  |           |
|-----------|--------------------------------------------------------------------------------------------------|-----------|
| 7(b)(i)   | (A method where the) reading (on the galvanometer) is zero.                                      | <b>B1</b> |
| 7(b)(ii)  | e.m.f. / 1.2 = 64 / 150                                                                          | <b>C1</b> |
|           | e.m.f. = (64 / 150) × 1.2<br>= 0.51 V                                                            | <b>A1</b> |
| 7(b)(iii) | (the internal resistance will cause a) drop in p.d. across the wire / the terminal p.d. is lower | <b>B1</b> |
|           | So the null point will move to the right                                                         | <b>B1</b> |

**11. M/J 25/P24/Q6/a,b,c**

|          |                                                                                 |             |
|----------|---------------------------------------------------------------------------------|-------------|
| 6(a)     | $2.4 / 6.0 = 8 / (R + 8)$                                                       | <b>C1</b>   |
|          | $R = 12 \Omega$                                                                 | <b>A1</b>   |
|          | or<br>$I = 2.4 / 8$<br>$= 0.30$<br>$R = (6 - 2.4) / 0.3$ or $R = (6 / 0.3) - 8$ | <b>(C1)</b> |
|          | $R = 12 \Omega$                                                                 | <b>(A1)</b> |
| 6(b)     | $V = I(\rho L / A)$ or $V = IR$ and $R = (\rho L / A)$                          | <b>M1</b>   |
|          | $I, \rho, A$ are constant (so $V \propto L$ )                                   | <b>A1</b>   |
| 6(c)(i)  | $V_{XP} / V_{XY} = L_{XP} / L_{XY}$<br>$E / 2.4 = 1.24 / 2.00$                  | <b>C1</b>   |
|          | $E = 1.5 \text{ V}$                                                             | <b>A1</b>   |
| 6(c)(ii) | p.d. across XY / wire increases / p.d. across XP increases                      | <b>M1</b>   |
|          | so P moved towards X / away from Y / to the left                                | <b>A1</b>   |

**12. O/N 24/P21/Q7/a,b**

|           |                                                                             |           |
|-----------|-----------------------------------------------------------------------------|-----------|
| 7(a)(i)   | $(V =) 1.50 - 1.36$<br>$(= 0.14 \text{ V})$                                 | <b>B1</b> |
|           | current = $V / R$<br>$= 0.14 / 0.28 = 0.50 \text{ A}$                       | <b>A1</b> |
| 7(a)(ii)  | resistance = $V / I$                                                        | <b>C1</b> |
|           | $= 1.36 / 0.50$                                                             | <b>A1</b> |
|           | $= 2.7 \Omega$                                                              |           |
| 7(a)(iii) | $R = 2.7 - 1.0$                                                             | <b>A1</b> |
|           | $= 1.7 \Omega$                                                              |           |
| 7(b)(i)   | two resistors correctly shown in parallel with cell and no other components | <b>B1</b> |

|          |                                                                                                                                                                                                                |           |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 7(b)(ii) | (external) resistance is now smaller<br><b>and</b><br>(so) current (in cell) is greater / (external) resistance smaller fraction of total resistance / internal resistance larger fraction of total resistance | <b>B1</b> |
|          | (greater p.d. across internal resistance so) terminal p.d. is less                                                                                                                                             | <b>B1</b> |

**13. O/N 24/P22/Q5/b(i,iii,iv,vii)**

|           |                                                        |             |
|-----------|--------------------------------------------------------|-------------|
| 5(b)(i)   | heater                                                 | <b>A1</b>   |
| 5(b)(iii) | $I = (230 - 6.0) / 170$<br>( = 1.3 A )                 | <b>C1</b>   |
|           | $I_1 = 7.0 - 1.3$<br>= 5.7 A                           | <b>A1</b>   |
| 5(b)(iv)  | $V = 230 - 6.0 - (5.7 \times 2.4)$                     | <b>C1</b>   |
|           | = 210 V                                                | <b>A1</b>   |
|           | <b>or</b>                                              |             |
|           | $R = ((230 - 6.0) / 5.7) - 2.4$<br>( = 36.9 $\Omega$ ) | <b>(C1)</b> |
|           | $V = 5.7 \times 36.9$<br>= 210 V                       | <b>(A1)</b> |
| 5(b)(vii) | (current) decreases                                    | <b>A1</b>   |

**14. O/N 24/P23/Q6/c**

|          |                                                                                                                                                                                                                                               |           |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 6(c)(i)  | correct symbols for resistor and thermistor, shown correctly connected in series with the battery                                                                                                                                             | <b>B1</b> |
| 6(c)(ii) | (as temperature increases) resistance of thermistor decreases                                                                                                                                                                                 | <b>M1</b> |
|          | (total resistance decreases so) greater current (in circuit/wire) so power (dissipated in the wire) increases<br><b>or</b><br>(total resistance decreases so) greater (share of) p.d. across wire so power (dissipated in the wire) increases | <b>A1</b> |

**15. M/J 24/P21/Q6/a,b,c,d**

|      |                                                                                                                                                                          |           |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 6(a) | <u>sum of</u> current(s) entering a junction = <u>sum of</u> current(s) leaving (the same junction)<br><b>or</b><br>(algebraic) sum of current (s) at a junction is zero | <b>B1</b> |
| 6(b) | (by Kirchhoff's second law) $V = V_1 + V_2$                                                                                                                              | <b>B1</b> |
|      | so $IR_T = IR_1 + IR_2$ (and cancelling $I$ gives) $R_T = R_1 + R_2$<br><b>or</b><br>$V/I = V_1/I + V_2/I$ (and substituting $R$ gives) $R_T = R_1 + R_2$                | <b>B1</b> |

|           |                                                                                          |             |
|-----------|------------------------------------------------------------------------------------------|-------------|
| 6(c)      | current in circuit = $1.35 / (10 + 15)$ (= 0.054 A)                                      | <b>C1</b>   |
|           | $r = (E - V) / I$                                                                        | <b>C1</b>   |
|           | = $(1.5 - 1.35) / 0.054$                                                                 | <b>A1</b>   |
|           | = $2.8 \Omega$                                                                           |             |
|           | <b>or</b>                                                                                |             |
|           | by potential divider principle $\frac{0.15}{1.35} = \frac{r}{25}$                        | <b>(C2)</b> |
|           | $r = 2.8 \Omega$                                                                         | <b>(A1)</b> |
|           | <b>or</b>                                                                                |             |
|           | $I = 1.35 / (10 + 15)$ (= 0.054 A)                                                       | <b>(C1)</b> |
| 6(d)(i)   | total resistance = $1.50 / 0.054$ (= $27.8 \Omega$ )                                     | <b>(C1)</b> |
|           | $r = 27.8 - 25$                                                                          |             |
|           | $r = 2.8 \Omega$                                                                         | <b>(A1)</b> |
| 6(d)(ii)  | the (total) resistance (of the circuit) has decreased (and e.m.f. is unchanged)          | <b>M1</b>   |
|           | (the current (in the cell) will) increase                                                | <b>A1</b>   |
| 6(d)(iii) | (as the current is greater and so there is a) larger p.d. across the internal resistance | <b>M1</b>   |
|           | (terminal p.d. will) decrease                                                            | <b>A1</b>   |

**16. M/J 24/P22/Q5/a,b**

|           |                                                                                        |           |
|-----------|----------------------------------------------------------------------------------------|-----------|
| 5(a)      | sum of electromotive force(s) = sum of potential difference(s) around a (closed) loop. | <b>B1</b> |
| 5(b)(i)   | $R = V / I$                                                                            | <b>C1</b> |
|           | = $(9.0 - 4.0) / 1.1 \times 10^{-2}$                                                   | <b>A1</b> |
|           | = $450 \Omega$                                                                         |           |
| 5(b)(ii)  | resistance (of thermistor) = $25 \Omega$ (from graph)                                  | <b>C1</b> |
|           | $V = E \times R_Y / (R_Y + R_X)$                                                       | <b>C1</b> |
|           | = $9 \times 25 / (25 + 450)$                                                           |           |
|           | <b>or</b>                                                                              |           |
|           | $I = E / R_{\text{Total}} = 9 / (25 + 450) = 1.89 \times 10^{-2} \text{ A}$            |           |
| 5(b)(iii) | $V = IR = 1.89 \times 10^{-2} \times 25$                                               |           |
|           | <b>or</b>                                                                              |           |
|           | $V_{(X)} = 9 \times 450 / (25 + 450) = 8.53 \text{ V}$                                 |           |
|           | $V = 9 - 8.53$                                                                         |           |
|           | $V = 0.47 \text{ V}$                                                                   | <b>A1</b> |
| 5(b)(iii) | (resistance of X increases so) the <u>total</u> resistance increases                   | <b>B1</b> |
|           | current decreases                                                                      | <b>B1</b> |
|           | potential difference (across thermistor / Y) decreases                                 | <b>B1</b> |

**17. M/J 24/P23/Q5/a,b,c**

|          |                                                                                                                                                                                       |           |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 5(a)(i)  | sum of electromotive force(s) = sum of potential difference(s) around a (closed) loop<br><b>or</b><br>the (algebraic) sum of the p.d.(s) and e.m.f.(s) is zero around a (closed) loop | <b>B1</b> |
| 5(a)(ii) | (law of conservation of) energy                                                                                                                                                       | <b>B1</b> |

|          |                                                                                   |             |
|----------|-----------------------------------------------------------------------------------|-------------|
| 5(b)(i)  | (by Kirchhoff's first law) $I = I_1 + I_2$                                        | <b>B1</b>   |
|          | $V/R_T = V/R_1 + V/R_2$ therefore $1/R_T = 1/R_1 + 1/R_2$                         | <b>B1</b>   |
| 5(b)(ii) | resistance of parallel combination = $(15 \times 10)/(15 + 10)$ (= 6.0 $\Omega$ ) | <b>C1</b>   |
|          | $r = (E - V)/I$                                                                   | <b>C1</b>   |
|          | $I = 1.38/6.0 = 0.23$ A                                                           | <b>A1</b>   |
|          | $r = (1.50 - 1.38)/0.23$                                                          |             |
|          | = 0.52 $\Omega$                                                                   |             |
|          | <b>or</b>                                                                         |             |
|          | (by potential divider principle) $r/R_T = Ir/V$                                   | <b>(C1)</b> |
|          | $r/6.0 = 0.12/1.38$                                                               | <b>(A1)</b> |
|          | $r = 0.52 \Omega$                                                                 |             |
|          | <b>or</b>                                                                         |             |
|          | (by potential divider equation) $V = E \times R_T/(R_T + r)$                      | <b>(C1)</b> |
|          | $1.38 = 1.5 \times 6.0/(6.0 + r)$                                                 | <b>(A1)</b> |
|          | $r = 0.52 \Omega$                                                                 |             |
|          |                                                                                   |             |
| 5(c)(i)  | as the (total) resistance has decreased (and e.m.f. is unchanged)                 | <b>M1</b>   |
|          | current will (in the cell) increase                                               | <b>A1</b>   |
| 5(c)(ii) | (as greater current means a) bigger drop in p.d. across the internal resistance   | <b>M1</b>   |
|          | p.d. (on voltmeter) will decrease                                                 | <b>A1</b>   |

**18. MAR 24/P22/Q7/b,c(i)**

|         |                                                                 |             |
|---------|-----------------------------------------------------------------|-------------|
| 7(b)    | $V = 0.25 \times 6$<br>= 1.5                                    | <b>C1</b>   |
|         | $Ir = E - IR$<br>$Ir = 1.8 - 1.5$<br>= 0.3                      | <b>C1</b>   |
|         | $r = 0.3/0.25$<br>= 1.2 $\Omega$                                | <b>A1</b>   |
|         | <b>or</b><br>(Total) $R = 1.8/0.25$<br>= 7.2<br>$E/I = (R + r)$ | <b>(C1)</b> |
|         | $1.8/0.25 = 6 + r$                                              | <b>(C1)</b> |
|         | $r = 7.2 - 6$<br>= 1.2 $\Omega$                                 | <b>(A1)</b> |
| 7(c)(i) | The same                                                        | <b>B1</b>   |

**19. O/N 23/P21/Q6/a(ii),b**

|          |                                                                                                                                                                                                                                                                                              |             |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| 6(a)(ii) | $R = (25 - 16) / 0.050$<br><b>or</b><br>$R = (9 / 16) \times 320$<br><b>or</b><br>$R = (25 / 0.050) - 320$                                                                                                                                                                                   | <b>C1</b>   |
|          | $R = 180 \, (\Omega)$                                                                                                                                                                                                                                                                        | <b>C1</b>   |
|          | $R_{(LDR)} = [(1 / 180) - (1 / 240)]^{-1}$<br>$= 720 \, \Omega$                                                                                                                                                                                                                              | <b>A1</b>   |
|          | <b>or</b>                                                                                                                                                                                                                                                                                    |             |
|          | $I = (25 - 16) / 240$<br>$(= 0.0375 \, \text{A})$                                                                                                                                                                                                                                            | <b>(C1)</b> |
|          | $I_{(LDR)} = 0.050 - 0.0375$<br>$(= 0.0125 \, \text{A})$                                                                                                                                                                                                                                     | <b>(C1)</b> |
|          | $R_{(LDR)} = 9.0 / 0.0125$<br>$= 720 \, \Omega$                                                                                                                                                                                                                                              | <b>(A1)</b> |
| 6(b)     | resistance of LDR decreases                                                                                                                                                                                                                                                                  | <b>B1</b>   |
|          | resistance of parallel combination decreases<br><b>or</b><br>total resistance (of circuit) decreases<br><b>or</b><br>current in resistor of resistance $320 \, \Omega$ increases<br><b>or</b><br>potential difference across parallel combination / LDR / $240 \, \Omega$ resistor decreases | <b>M1</b>   |
|          | voltmeter reading increases                                                                                                                                                                                                                                                                  | <b>A1</b>   |
|          |                                                                                                                                                                                                                                                                                              |             |

**20. O/N 23/P22/Q6/b(iii),c(ii,iii)**

|           |                                                                                                                                                      |           |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 6(b)(iii) | $6.4 / E = 1600 / (1400 + 1600)$<br><b>or</b><br>$E = 6.4 + (4.0 \times 10^{-3} \times 1400)$<br><b>or</b><br>$E = 4.0 \times 10^{-3} (1600 + 1400)$ | <b>C1</b> |
|           | $E = 12 \, \text{V}$                                                                                                                                 | <b>A1</b> |
| 6(c)(ii)  | increase                                                                                                                                             | <b>B1</b> |
| 6(c)(iii) | decrease                                                                                                                                             | <b>B1</b> |

**21. O/N 23/P23/Q7/b(ii,iii,iv)**

|           |                                                                                                                                |           |
|-----------|--------------------------------------------------------------------------------------------------------------------------------|-----------|
| 7(b)(ii)  | voltmeter reading = $3.1 \, \text{V}$                                                                                          | <b>A1</b> |
| 7(b)(iii) | current in the battery: increase                                                                                               | <b>B1</b> |
|           | voltmeter reading: decrease                                                                                                    | <b>B1</b> |
| 7(b)(iv)  | cross marked on the resistance wire to right of the arrowhead of S, but not touching the right-hand end of the resistance wire | <b>B1</b> |

**22. MAR 23/P22/Q6**

- (a) energy (transferred from electrical to other forms) per unit charge [1]
- (b) (i) (resistance is) infinite / very high [1]  
 (ii) (resistance) decreases (as  $V$  increases) [1]
- (c) (i) current =  $2.7 - 1.5$   
           =  $1.2 \text{ A}$  [1]
- (ii)  $12 = (1.5 \times 5.0) + (1.5 \times R)$  or  $R = (12/1.5) - 5.0$   
        $R = 3.0 \Omega$  [2]
- (iii)  $V_{(XZ)} = (1.6/2.0) \times 12 (= 9.6 \text{ V})$   
        $V_{(XW)} = 1.5 \times 5.0 (= 7.5 \text{ V})$   
       potential difference =  $9.6 - 7.5$   
                                   =  $2.1 \text{ V}$   
       or  
        $V_{(ZY)} = (0.4/2.0) \times 12 (= 2.4 \text{ V})$   
        $V_{(WY)} = 1.5 \times 3.0 (= 4.5 \text{ V})$   
       potential difference =  $4.5 - 2.4$   
                                   =  $2.1 \text{ V}$  [3]
- (iv) current in (fixed / variable) resistor decreases [3]  
       current in (resistance) wire is unchanged  
       (so) current in battery decreases, (same e.m.f. so) power decreases [3]

**23. O/N 22/P22/Q6**

- (a) energy transferred per (unit) charge (from electrical to other forms) [1]
- (b) same/equal current (in X and Y)  
        $P = I^2 R$  (and  $R_X > R_Y$ )  
       or  
        $P = VI$  and  $V_X > V_Y$   
       (so) X (dissipates more power) [3]
- (c) (i)  $I = Q/t$   
       =  $650/540$   
       =  $1.2 \text{ A}$  [2]
- (ii)  $V = W/Q$  or  $W/It$   
       =  $4800/650$  or  $4800/(1.2 \times 540)$   
       =  $7.4 \text{ V}$   
       or  
        $V = P/I$  and  $P = W/t$   
       =  $8.9/1.2$   
       =  $7.4 \text{ V}$  [2]
- (iii)  $I = 4.5 + 1.2 (= 5.7 \text{ A})$   
        $9.0 = 7.4 + 5.7r$  or  $9.0 = 5.7(1.3 + r)$   
        $r = 0.28 \Omega$  [2]

**24. O/N 22/P23/Q5****(a)** sum of e.m.f.(s) = sum of p.d.(s)**or**(algebraic) sum of e.m.f.(s) and p.d.(s) is zero  
around a loop / around a closed circuit

[2]

**(b) (i)**  $1/r_{(T)} = 1/0.59 + 1/0.59 + 1/0.59$ 

$$(r_{(T)}) = 0.197 \, (\Omega)$$

$$(R =) 2.2 - 0.197 = 2.0 \, \Omega$$

**or** $I = 1.5/2.2 (= 0.68 \, \text{A})$  **and**  $i = 0.68/3$  (where  $I$  is the circuit current **and**  $i$  is the current from each cell)

$$(E = IR + ir =) 1.5 = 0.68R + (0.68/3) \times 0.59 \text{ **and** } R = 2.0 \, \Omega$$

[2]

**(ii)** current =  $1.5/2.2$ 

$$= 0.68 \, \text{A}$$

**or**

p.d. across cell = p.d. across conductor

$$1.5 - 0.59I = 3I \times 2.0 \text{ so } I = 0.228 \, \text{A (where } I \text{ is current in cell)}$$

$$\text{current} = 3 \times 0.228$$

$$= 0.68 \, \text{A}$$

**or**current in conductor =  $3 \times$  current in cell

$$V/2.0 = 3 \times (1.5 - V)/0.59 \text{ (where } V \text{ is p.d. across conductor)}$$

$$V = 1.37 \, \text{V}$$

$$\text{current} = 1.37/2.0$$

$$= 0.68 \, \text{A}$$

[2]

**(c) (i)**  $R = \rho L / A$ 

$$R = 4\rho L / \pi d^2$$

( $\rho$  and  $L$  are the same so)

$$R_A / R_B = 7.6^2 / 4.3^2$$

$$= 3.1$$

[3]

**(ii)**  $I = Anvq$  **and**  $I$ ,  $n$ ,  $q$  are same / equal / constant

$$\frac{V_A}{V_B} = \frac{A_B}{A_A} = \frac{d_B^2}{d_A^2}$$

$$\text{ratio} = 7.6^2 / 4.3^2$$

$$= 3.1$$

[2]

**(d)** combined internal resistance (of the cells) will be greater**or**

total / circuit resistance (of circuit) greater (because a parallel resistance removed)

more 'lost volts' (inside each cell) **or** internal resistances take a greater share of total p.d.**or** conductor gets a smaller shareof the total p.d. **or** current in conductor/total current decreases

(so) potential difference (across conductor) decreases

[3]

**25. M/J 22/P22/Q6**

**(a)**  $V = V_1 + V_2 + V_3$

$$IR = IR_1 + IR_2 + IR_3 \text{ **or** } (V/I) = (V_1/I) + (V_2/I) + (V_3/I) \text{ **and** }$$

$$R = R_1 + R_2 + R_3$$

[2]

- (b) (i)  $V/8.0 = 6.0 \times 10^3 / (4.0 \times 10^3 + 6.0 \times 10^3)$   
or  
 $I = 8.0 / (4.0 \times 10^3 + 6.0 \times 10^3) = 8.0 \times 10^{-4}$   
 $V = 8.0 \times 10^{-4} \times 6.0 \times 10^3$   
 $V = 4.8 \text{ V}$  [2]
- (ii) total resistance in parallel  $= 3.0 \times 10^3 (\Omega)$  or  $3.0 (\text{k}\Omega)$   
current  $= 8.0 / (3.0 \times 10^3 + 6.0 \times 10^3)$   
 $= 8.9 \times 10^{-4} \text{ A}$  [2]
- (c) thermistor resistance increases  
(thermistor resistance increases so total resistance increases so) current decreases  
(in battery)  
( $P = EI$  and  $E$  constant so) power decreases [3]

## 26. M/J 22/P21/Q6

- (a) sum of current(s) into junction = sum of current(s) out of junction  
or  
(algebraic) sum of current(s) at a junction is zero [1]
- (b) (i) same potential difference (across X and Y as in parallel)  
power  $= V^2/R$  (and  $R_X > R_Y$ )  
or  
power  $= VI$  and  $I_X < I_Y$   
(so) Y (dissipates more power) [3]
- (ii)  $R = \rho L/A$  (and  $R_X > R_Y$ )  
(so) Y (has the larger (cross-sectional) area) [2]
- (c) (i) current  $= 5.4/1800$   
 $= 3.0 \times 10^{-3} \text{ A}$  [1]
- (ii)  $5.4/9.0 = 1800/(1800 + R_L)$   
or  
 $R_L = (9.0 - 5.4)/3.0 \times 10^{-3}$   
 $R_L = 1200 \Omega$  [2]
- (iii) resistance of LDR  $/ R_L$  decreases  
current (in the circuit) increases (so) voltmeter reading increases [2]

## 27. O/N 21/P22/Q6

- (a)  $Q = It$   
 $= 0.80 \times 7.5 \times 60$   
 $= 360 \text{ C}$  [2]
- (b)  $P = EI$  or  $P = VI$  or  $P = I^2R$  or  $P = V^2/R$   
 $0.80^2 \times 0.40 (= 0.256 \text{ W})$   
or  
 $0.48 \times 0.80 (= 0.384 \text{ W})$   
efficiency  $= (0.256/0.384) \times 100$   
 $= 67\%$  [3]
- (c) (i)  $n = 3.2 \times 10^{22} / (1.3 \times 10^{-7} \times 3.0)$   
 $= 8.2 \times 10^{28} \text{ m}^{-3}$  [1]

(ii)  $I = Anvq$

$$v = 0.80 / (1.3 \times 10^{-7} \times 8.2 \times 10^{28} \times 1.60 \times 10^{-19})$$

$$= 4.7 \times 10^{-4} \text{ m s}^{-1}$$

[2]

- (d) (wire Y has) larger resistance / resistance increases  
 (wire Y has) smaller current / current decreases  
 (average drift) speed is less (in wire Y)

[3]

## 28. O/N 21/P21/Q5

- (a) sum of current(s) in = sum of current(s) out  
 or

(algebraic) sum of current(s) is zero  
 at a junction (in a circuit)

[2]

- (b) (i) (current in  $R_4$  or  $R_1$ )  $= 0.30 + 0.30$   
 $(= 0.60 \text{ A})$

$$(R =) 2.4 / 0.60 = 4.0 (\Omega)$$

or

$$(\text{p.d. across } R_3 \text{ or } R_2 =) 2.4 / 2$$

$$(= 1.2 \text{ V})$$

$$(R =) 1.2 / 0.30 = 4.0 (\Omega)$$

[2]

- (ii)  $E = 2.4 + 2.4 + 1.2$   
 $= 6.0 \text{ V}$

or

$$\text{total resistance} = 10 (\Omega)$$

$$E = 10 \times 0.60$$

$$= 6.0 \text{ V}$$

[2]

- (c) total resistance increases  
 current decreases (in battery) so total power decreases

[2]

- (d) resistivity  $= RA / L$   
 $= 4.0 \times \pi \times (240 \times 10^{-6})^2 / 0.67$   
 $= 1.1 \times 10^{-6} \Omega \text{ m}$

[3]

## 29. O/N 21/P23/Q6

(a)  $R = \rho L / A$

$$(R =) 5(.0) \times 10^{-7} \times 2(.0) / 3.3 \times 10^{-7} = 3.0 \Omega$$

[2]

(b) (i)  $I = 1.2 / 3.0$   
 $= 0.40 \text{ A}$

[1]

(ii)  $r = (1.50 - 1.20) / 0.40$  or  $1.50 / 0.40 - 3.0$   
 $= 0.75 \Omega$

[2]

(c)  $E / 1.20 = 1.4 / 2.0$   
 $E = 0.84 \text{ V}$

or

$$R_{XP} = (1.4 / 2.0) \times 3.0 (= 2.1 \Omega)$$

$$E = 2.1 \times 0.40$$

$$E = 0.84 \text{ V}$$

[2]

- (d) (second wire has) larger resistance/resistance increases  
 p.d. across XY is larger/increases (for second wire)  
 or  
 p.d. across the (second) wire is larger/increases  
 (so) length XP (for second wire) is shorter

[3]

**30. M/J 21/P21/Q5**(a) sum of e.m.f.(s) = sum of p.d.(s)

or

(algebraic) sum of e.m.f.(s) and p.d.(s) is zero  
around a loop/around a closed circuit

[2]

(b) (i)  $I = 1.8 / 0.90$   
 $= 2.0 \text{ A}$

[1]

(ii)  $Q = It$

number  $= (2.0 \times 45) / 1.60 \times 10^{-19}$   
 $= 5.6 \times 10^{20}$

[2]

(iii)  $4.0 = 1.8 + [2.0 \times (0.35 + R)]$

or

$4.0 = 2.0 \times (0.90 + 0.35 + R)$

$R = 0.75 \Omega$

[2]

(c) (i)  $1.2 / 1.8 = 0.30 / L$

$L = 0.45 \text{ m}$

[2]

(ii) p.d. across XY decreases/p.d. across XP decreases  
(so) P is moved towards Y/away from X/to the right

— [2]

**31. MAR 21/P22/Q6/b(iv)**

(a) (i)  $I = Anvq$

ratio  $= (360 \text{ A} / A) \times (2.5 \text{ n} / n)$  or  $360 \times 2.5$   
 $= 900$

[2]

**32. O/N 20/P22/Q6**

(a)  $I = I_1 + I_2 + I_3$

$(V/R) = (V/R_1) + (V/R_2) + (V/R_3)$  or  $(I/V) = (I_1/V) + (I_2/V) + (I_3/V)$

and

$1/R = 1/R_1 + 1/R_2 + 1/R_3$

[2]

(b) (i) current  $= 0.49 + 0.45$   
 $= 0.94 \text{ A}$

[1]

(ii)  $8.0 = (0.94 \times r) + (0.45 \times 16)$   
 $r = 0.85 \Omega$

[2]

(c)  $I = Anvq$

$v = (0.45 / 0.49) \times 2.1 \times 10^{-4}$   
 $= 1.9 \times 10^{-4} \text{ m s}^{-1}$

[2]

(d) total/combined resistance decreases

(current in battery increases so terminal) potential difference decreases

[2]

**33. O/N 20/P23/Q6**

(a)  $\frac{\text{work (done) / energy (transferred from electrical to other forms)}}{\text{charge}}$

[1]

- (b)  $R = \rho L / A$  [2]  
 $V = LA$  and (so)  $R = \rho V / A^2$  (with  $\rho$  and  $V$  constant)
- (c)  $E = IR + Ir$  or  $E = I(R + r)$  or  $E - Ir = IR$  [1]  
 and  
 $R = (E / I) - r$
- (d) (i)  $P = I^2 R$  or  $P = IV$  or  $P = V^2 / R$   
 $R = 5.4 \, (\Omega)$  or  $V = 10.8 \, (\text{V})$   
 $P = 2.0^2 \times 5.4$  [3]  
 $= 22 \, \text{W}$
- (ii) 1.  $r = 0.60 \, \Omega$   
 2.  $E = \text{gradient}$   
 $= \text{e.g. } 5.4 / 0.45$  [3]  
 $= 12 \, \text{V}$

**34. M/J 20/P22/Q6**

- (a) (i) energy is dissipated in the internal resistance/ $r$  [1]  
 (ii) 1.  $I = Q / t$   
 $= 750 / 1500$   
 $= 0.50 \, \text{A}$  [2]
2.  $V = W / Q$  or  $V = W / It$   
 $= 5700 / 750$  or  $5700 / (0.50 \times 1500)$   
 $= 7.6 \, \text{V}$   
 or  
 $V = P / I$  and  $P = W / t$   
 $V = 3.8 / 0.50$  [2]  
 $= 7.6 \, \text{V}$
3.  $r = (7.8 - 7.6) / 0.50$  [2]  
 $= 0.40 \, \Omega$  [1]
- (b) (i)  $90 \, \Omega$  and  $45 \, \Omega$  resistors shown connected in parallel [1]  
 (ii) the resistors connected in parallel labelled as  $90 \, \Omega$  and  $45 \, \Omega$  with the other resistor labelled as  $20 \, \Omega$   $V_{\text{OUT}}$  or  $3.6 \, \text{V}$  labelled across the  $20 \, \Omega$  resistor [2]

**35. M/J 20/P23/Q6**

- (a)  $A$ : cross-sectional area [2]  
 $n$ : number density of free electrons
- (b) units of  $I$ :  $\text{A}$  and units of  $A$ :  $\text{m}^2$  and units of  $v$ :  $\text{m s}^{-1}$   
 units of  $e$ :  $\text{A} / (\text{m}^2 \text{m}^{-3} \text{m s}^{-1}) = \text{A s}$  [2]
- (c) ratio  $= A_Q / A_P$   
 $= [\pi r^2] / [\pi (2r^2)]$  [2]  
 $= 0.25$

**36. MAR 20/P22/Q5**

- (a) volt/ampere [1]  
 (b)  $R = \rho L / A$   
 $L = (1.8 \times 0.38 \times 10^{-6}) / 9.6 \times 10^{-7}$  [3]  
 $= 0.71 \, \text{m}$
- (c) (i) thermal energy is dissipated in resistor Y [1]

- (ii)  $V / 1.2 = 1.8 / (1.8 + 0.6)$   
 $V = 0.90 \text{ V}$   
 or  
 $I = 1.2 / (1.8 + 0.6) (= 0.50)$   
 $V = 0.50 \times 1.8$   
 $= 0.90 \text{ V}$  [2]
- (d) (i) remain the same [1]  
 (ii) decrease [1]
- (e) (i)  $1 / R = 1 / 1.8 + 1 / 3.6$  [1]  
 $R = 1.2 \Omega$
- (ii)  $I = 1.2 / (1.2 + 0.60)$  [1]  
 $= 0.67 \text{ A}$   
 or  
 $V_Y = 1.2 \times 0.60 / (1.2 + 0.60) (= 0.40)$   
 $I = 0.40 / 0.60$   
 $= 0.67 \text{ A}$  [2]

**37. O/N 19/P22/Q6**

- (a) sum of current(s) into junction = sum of current(s) out of junction  
 or  
 (algebraic) sum of current(s) at a junction is zero [1]
- (b) (i)  $R = V / I$   
 $= 0.60 / 7.5 \times 10^{-3}$   
 $= 80 \Omega$
- (ii) resistance decreases [3]
- (c) (i)  $E = 0.60 + 0.30$  [1]  
 $= 0.90 \text{ V}$
- (ii)  $(I =) 9.3 - 7.5$  [1]  
 $I = 1.8 \text{ (mA) or } 1.8 \times 10^{-3} \text{ (A)}$   
 $R = 0.90 / 1.8 \times 10^{-3}$   
 $= 500 \Omega$   
 or  
 total resistance  $= 0.90 / 9.3 \times 10^{-3} = 96.8 \text{ (}\Omega\text{)}$   
 total resistance of diode and X  $= 0.90 / 7.5 \times 10^{-3} = 120 \text{ (}\Omega\text{)}$   
 $1 / 96.8 = 1 / R + 1 / 120$   
 $R = 500 \Omega$
- (iii)  $P = VI \text{ or } I^2 R \text{ or } V^2 / R$  [2]  
 $= 0.60 \times 7.5 \times 10^{-3} \text{ or } (7.5 \times 10^{-3})^2 \times 80 \text{ or } 0.60^2 / 80$   
 $= 4.5 \times 10^{-3} \text{ W}$
- (iv) current  $= 2.5 \text{ mA}$  [2]  
 [1]

**38. O/N 19/P23/Q6**

- (a) (i)  $R = V / I$   
 resistance  $= (12 / 0.20) / 2 \text{ or } 6 / 0.20$   
 $= 30 \Omega$
- (ii)  $I = 0.50 - 0.20 (= 0.30 \text{ A})$  [2]  
 $R + 28 = 12 / 0.30 (= 40 \Omega)$   
 $R = 12 \Omega$  [2]

(b) p.d. across lamp =  $0.20 \times 30$  (= 6.0 V)

p.d. across  $R = 0.30 \times 12$  (= 3.6 V)

$$V_{XY} = 6.0 - 3.6$$

$$= 2.4 \text{ V}$$

or

p.d. across lamp =  $0.20 \times 30$  (= 6.0 V)

p.d. across  $28 \Omega$  resistor =  $0.30 \times 28$  (= 8.4 V)

$$V_{XY} = 8.4 - 6.0$$

$$= 2.4 \text{ V}$$

[3]

(c)  $P = VI$  or  $P = EI$  or  $P = I^2 R$  or  $P = V^2 / R$

$$\text{ratio} = (6.0 \times 0.20) \times 2 / (12 \times 0.50) \text{ or } 0.20 / 0.50$$

$$= 0.40$$

[2]

(d) no change to  $V$  across lamps, so power in lamps unchanged

or

current in battery/total current increases (and e.m.f. the same) so power produced by battery increases both the above statements and so the ratio decreases

[2]

### 39. M/J 19/P22/Q5

(a) sum of e.m.f.(s) = sum of p.d.(s)

around a loop/around a closed circuit

[2]

(b) (i) 1.  $1/R = 1/R_1 + 1/R_2$

$$1/R = 1/90 + 1/18$$

$$R = 15 \Omega$$

[2]

2.  $I = V/R$

$$I = 4.8/15 \text{ or } I = 4.8/90 + 4.8/18$$

$$I = 0.32 \text{ A}$$

[2]

(ii)  $E = V + Ir$

or

$$E = I(R + r)$$

$$5.6 = 4.8 + 0.32 r \text{ so } r = 2.5 (\Omega)$$

or

$$5.6 = 0.32 \times (15 + r) \text{ so } r = 2.5 (\Omega)$$

V [2]

[2]

(iii)  $P = EI$  or  $P = VI$  or  $P = I^2 R$  or  $P = V^2 / R$

$$\text{ratio} = (0.32^2 \times 2.5) / (5.6 \times 0.32) \text{ or } 0.256 / 1.792$$

$$= 0.14$$

[3]

(c)  $7.2 - 5.6 - 2.5I - 3.5I = 0$

$$I = 0.27 \text{ A}$$

[2]

[1]

### 40. M/J 19/P23/Q6

(a) volt/ampere

(b) (i) 1.  $I = 1.8 + 0.60$

$$= 2.4 \text{ A}$$

[1]

2.  $(8.0 \times 0.60) = 1.8 \times (2.0 + R_2)$

$$R_2 = 0.67 \Omega$$

[2]

3.  $E - (2.4 \times 1.5) = (0.60 \times 8.0)$

or

$$E - (2.4 \times 1.5) = 1.8 \times (2.0 + 0.67)$$

or

$$E = 2.4 \times [1.5 + (8.0 \times 2.67) / (8.0 + 2.67)]$$

$$E = 8.4 \text{ V}$$

[2]

$$(ii) \quad 1. \quad R = \rho L / A \quad \text{or} \quad R \propto 1 / A$$

$$\text{ratio} = R_Y / R_X = 2.0 / 8.0$$

$$= 0.25$$

[2]

$$2. \quad I \propto Av \quad \text{or} \quad I_X / I_Y = A_X v_X / A_Y v_Y$$

$$\text{ratio} = (0.60 / 1.8) \times (1 / 0.25)$$

$$= 1.3$$

[2]

**41. O/N 18/P22/Q6**

(a) joule/coulomb

[1]

$$(b) \quad (i) \quad 7.0 = (I \times 5.2) + (I \times 6.0) + 1.4$$

$$I = 0.50 \text{ A}$$

[2]

$$(ii) \quad R = 1.4 / 0.50$$

$$= 2.8 \Omega$$

[1]

$$(iii) \quad P = EI \quad \text{or} \quad P = VI \quad \text{or} \quad P = I^2 R \quad \text{or} \quad P = V^2 / R$$

$$\text{efficiency} = [(0.50^2 \times 6.0) / (7.0 \times 0.50)] (\times 100)$$

or

$$\text{efficiency} = [(0.50 \times 3.0) / (7.0 \times 0.50)] (\times 100)$$

or

$$\text{efficiency} = [(3.0^2 / 6.0) / (7.0 \times 0.50)] (\times 100)$$

$$\text{efficiency} = 43\%$$

[3]

$$(iv) \quad R = \rho l / A$$

$$\alpha = \rho / R$$

$$= 3.7 \times 10^{-7} / 6.0$$

$$= 6.2 \times 10^{-8} \text{ m}$$

[2]

**42. O/N 18/P21/Q6**(a) sum of e.m.f.(s) equal to sum of p.d.(s)

around a loop/around a closed circuit

[2]

$$(b) \quad (i) \quad \text{current in variable resistor} = (6.0 / 2.4) + (6.0 / 1.2) (= 7.5 \text{ A})$$

$$\text{p.d. across variable resistor} = 9.0 - 6.0 (= 3.0 \text{ V})$$

$$R = 3.0 / 7.5$$

$$= 0.40 \Omega$$

or

$$\frac{1}{R_T} = \frac{1}{2.4} + \frac{1}{1.2}$$

$$R_T = 0.80 (\Omega)$$

$$\frac{3}{9} = \frac{R}{(R + 0.80)} \quad \text{or} \quad \frac{3}{R} = \frac{6}{0.8}$$

$$R = 0.40 \Omega$$

[3]

$$(ii) \quad P = V^2 / R \quad \text{or} \quad P = I^2 R \quad \text{or} \quad P = IV$$

$$P = 6.0^2 / 24 \quad \text{or} \quad 2.5^2 \times 2.4 \quad \text{or} \quad 6.0 \times 2.5$$

$$= 15 \text{ W}$$

[2]

- (iii) 1.  $R = \frac{\rho L}{A}$   
 ratio =  $(2.4/1.2) \times (3/1)$   
 = 6.0 [2]
2. ( $I = nAvq$ )  
 $I_X/I_Y = 2.5/5.0$  or  $1.2/2.4$  or 0.5  
 ratio =  $(2.5/5.0) \times (1/3)$  or  $(1.2/2.4) \times (1/3)$   
 = 0.17 [2]

## 43. O/N 18/P23/Q7

- (a) sum of current(s) in(to) junction = sum of current(s) out of junction  
 or  
 (algebraic) sum of current(s) at a junction is zero [1]
- (b) (i) 1. potential difference = 0  
 2. potential difference = 9.6V [2]
- (ii) for resistance in parallel:  $(1/R_T) = (1/400) + (1/400)$   
 $R_T = 200(\Omega)$   
 $V/9.6 = 200/600$   
 $V = 3.2V$  [3]

## 44. M/J 18/P22/Q6

- (a) (i) sum of current(s) into junction = sum of current(s) out of junction  
 or  
 (algebraic) sum of current(s) at a junction is zero [1]  
 (ii) charge [1]
- (b) (i) 1.  $E = I^2 R t$  or  $E = V I t$  or  $E = (V^2 / R) t$   
 $E = 2.5^2 \times 2.0 \times 5.0 \times 60$  or  $5.0 \times 2.5 \times 5.0 \times 60$  or  $(5.0^2 / 2.0) \times 5.0 \times 60$   
 = 3800J [2]
2. p.d. =  $8.0 - (2.0 \times 2.5)$   
 = 3.0V [1]
- (ii)  $I_X = 3.0/15 = 0.20$  (A)  
 $I_Y = 2.5 - 0.20 = 2.3$  (A)  
 $R_Y = 3.0/2.3$   
 = 1.3  $\Omega$   
 or  
 $R_T = 3.0/2.5 = 1.2(\Omega)$  or  $(8.0/2.5) - 2.0 = 1.2(\Omega)$   
 $1/1.2 = 1/15 + 1/R_Y$   
 $R_Y = 1.3\Omega$  [3]
- (iii) 1. Z has larger radius/diameter/(cross-sectional) area  
 Z has (material of) smaller resistivity/greater conductivity [2]
2. current/I (in battery) increases  
 ( $P = EI$  so) power/P (produced by battery) increases [2]

## 45. M/J 18/P23/Q6

- (a)  $R = \rho L / A$   
 $3.0 = \rho / [\pi \times (0.48 \times 10^{-3} / 2)^2]$   
 $\rho = 5.4 \times 10^{-7} \Omega m$  [3]

- (b) (i) p.d. =  $5.0 - (2.0 \times 1.6)$   
 $= 1.8\text{V}$  [1]
- (ii) 1. current in resistor =  $1.8/4.5 (= 0.40\text{A})$   
 current in wire =  $1.6 - 0.40 (= 1.2\text{A})$   
 $R_x = 1.8/1.2$   
 $= 1.5\Omega$   
 or  
 $R_T = 1.8/1.6$  or  $(5.0/1.6) - 2.0 (= 1.125\Omega)$   
 $(1/1.125) = (1/4.5) + (1/R_x)$   
 $R_x = 1.5\Omega$  [3]
2. length =  $1.5/3.0$  or  $1.5 \times 1.8 \times 10^{-7} / (5.4 \times 10^{-7})$   
 $= 0.50\text{m}$  [1]

**46. MAR 18/P22/Q5**

- (a) sum of e.m.f.(s) = sum of p.d.(s)  
 around a loop / around a closed circuit [2]
- (b) (i) 1  $6.0 - 4.0I = 0$   
 $I = 1.5\text{A}$  [1]
- 2  $6.0 + 6.0 = I(4.0 + R + 1.5)$   
 $12 = 1.5(4.0 + R + 1.5)$   
 $R = 2.5\Omega$   
 or  $6.0 = I(R + 1.5)$   
 $6.0 = 1.5(R + 1.5)$   
 $R = 2.5\Omega$   
 or combines  $6 = 4I$  and  $6 = I(R + 1.5)$  to give  
 $4 = R + 1.5$   
 $R = 2.5\Omega$  [2]
- (ii)  $I = Anvq$  [2]  
 ratio =  $1^2/2^2$   
 $= 0.25$  [2]
- (iii) total (circuit) resistance increases [2]  
 current /  $I$  decreases or  $P \propto I$  or  $P \propto 1 / (\text{total resistance})$   
 power (transformed) decreases [3]

**47. O/N 17/P21/Q5**

- (a)  $I_1 + I_2 = I_3$  [any subject] [1]
- (b)  $E_1 + E_3 = I_1R_1 + I_3R_3 + I_3R_4$  [any subject] [1]
- (c)  $E_1 - E_2 = I_1R_1 - I_2R_2$  [any subject] [1]

**48. M/J 17/P22/Q7**

- (a) energy transformed from chemical to electrical / unit charge (driven around a complete circuit) [1]
- (b) (i) the current decreases (as resistance of Y increases)  
 lost volts go down (as resistance of Y increases)  
 p.d. AB increases (as resistance of Y increases) [3]
- (ii) 1.  $1.50 = 0.180 \times (6.00 + 0.200 + R_x)$   
 $R_x = 2.1(3)\Omega$  [2]

2. p.d. AB =  $1.5 - (0.180 \times 0.200)$  or  $0.18 \times (2.13 + 6.00)$   
 = 1.46(4) V [2]

3. efficiency = (useful) power output / (total) power input or  $IV/IE$   
 (=  $1.46 / 1.5$ ) = 0.97 [0.98 if full figures used] [2]

**49. O/N 16/P22/Q5**

(a) total/sum of electromotive forces or e.m.f.s  
= total/sum of potential differences or p.d.s  
around a loop/(closed) circuit [2]

(b) (i) (current in battery =) current in A + current in B or  $I_A + I_B$  [2]  
( $I =$ )  $0.14 + 0.26 = 0.40\text{A}$

$$\begin{aligned} \text{(ii)} \quad E &= V + Ir \\ 6.8 &= 6.0 + 0.40r \quad \text{or} \quad 6.8 = 0.40(15 + r) \\ r &= 2.0 \Omega \end{aligned} \quad [2]$$

(iii)  $R = V/I$   
ratio ( $= R_A/R_B$ )  $= (6.0/0.14)/(6.0/0.26)$   
 $= 42.9/23.1$  or  $0.26/0.14$   
 $= 1.9$  (1.86) [2]

$$\text{(iv) 1. } P = EI \text{ or } VI \quad \text{or} \quad P = I^2 R \quad \text{or} \quad P = V^2 / R$$

$$= 6.8 \times 0.40 \quad = 0.40^2 \times 17 \quad = 6.8^2 / 17$$

$$= 2.7 \text{ W (2.72 W)} \quad [2]$$

2. output power =  $VI$   
 $= 6.0 \times 0.40 (= 2.40 \text{ W})$   
 efficiency =  $(6.0 \times 0.40) / (6.8 \times 0.40) = 2.40 / 2.72$   
 $= 0.88 \text{ or } 88\% \text{ (allow } 0.89 \text{ or } 89\%)$  [2]

(c) charge exists only in discrete amounts [1]

**50. M/J 16/P22/Q7**

**(a) charge exists only in discrete amounts**

(b) (i)  $E = I(R + r)$  or  $V = IR$   
(total resistance =)  $2.7 + 0.30 + 0.25 (= 3.25 \Omega)$   
 $I = 9.0 / (2.7 + 0.30 + 0.25)$  or  $9.0 / 3.25 = 2.8 \text{ A}$  [3]

$$\begin{aligned} \text{(ii)} \quad V &= IR_{\text{ext}} \\ &= 2.77 \times 3.0 \quad \text{or} \quad 2.8 \times 3.0 \\ \text{or} \\ V &= E - Ir \\ &= 9.0 - 2.77 \times 0.25 \quad \text{or} \quad 9.0 - 2.8 \times 0.25 \\ V &= 8.3 \text{ (8.31) V} \quad \text{or} \quad 8.4 \text{ V} \end{aligned} \quad [2]$$

(c) (i)  $I = nevA$   
 $v = 2.77 / (8.5 \times 10^{29} \times 1.6 \times 10^{-19} \times 2.5 \times 10^{-6})$   
 $= 8.1 \text{ (8.147)} \times 10^{-6} \text{ ms}^{-1} \text{ or } 8.2 \times 10^{-6} \text{ ms}^{-1}$  [2]

(ii) A reduces by a factor 4 ( $1/4$  less) or resistance of Z goes up by  $4\times$   
current goes down but by less than a factor of 4 (as total resistance  
does not go up by a factor of 4) so drift speed goes up [2]

**51. O/N 15/P22/Q5**

(a) (i)  $I = V/R$   
 $(= 240/1500 =) 0.16 \text{ A}$  [2]

- (ii)  $I_2 = 0.40 - 0.16 (= 0.24)$   
 $0.24 (350 + R) = 240$   
 $R = 650 \Omega$  [2]
- (iii) power =  $IV$  or  $I^2 R$  or  $V^2/R$   
 ratio =  $(84 \times 0.24)/(88 \times 0.16)$   
 or  $[(0.24)^2 \times 350]/[(0.16)^2 \times 550]$   
 or  $(84^2/350)/(88^2/550)$   
 or  $20.16/14.08$   
 $= 1.4(3)$  [2]
- (b) (i) p.d. across  $350 \Omega$  resistor =  $0.24 \times 350$   
 or p.d. across  $550 \Omega$  resistor =  $0.16 \times 550$   
 $V_{350} = 84$  (V) and  $V_{550} = 88$  (V) gives  $V_{AB} = 4.0$  V  
 or  $V_{950} = 152$  (V) and  $V_R = 156$  V gives  $V_{AB} = 4.0$  V [2]
- (ii) p.d. across  $R$  increases or potential at B increases or  $V_{350}$  decreases hence  $V_{AB}$  increases [1]

## 52. O/N 14/P22/Q5

- (a) (i) in series  $2X$  or in parallel  $X/2$   
 other relationship given and  $4\times$  greater in series (than in parallel) [2]
- (ii) due to the internal resistance  
 total resistance for series circuit is not four times greater than resistance for parallel circuit [2]
- (iii) 1.  $E = I_1(2X + r)$  or  $12 = 1.2(2X + r)$   
 2.  $E = I_2(X/2 + r)$  or  $12 = 3.0(X/2 + r)$  [2]
- (iv)  $2X + r = 10$  and  $X/2 + r = 4$   
 $X = 4.0 \Omega$  [1]
- (b)  $P = I^2 R$  or  $V^2/R$  or  $VI$   
 ratio =  $[(1.2)^2 \times 4]/[(1.5)^2 \times 4]$   
 $= 0.64$  [2]
- (c) the resistance (of a lamp) changes with  $V$  or  $I$   
 $V$  or  $I$  is greater in parallel circuit or circuit 2  
 or  $V$  or  $I$  is less in series circuit or circuit 1 [2]

## 53. O/N 14/P21/Q7

- (a) due to the lost volts in internal resistance/cell or energy losses in the internal resistance/cell [1]
- (b) (i)  $V = IR$   
 $= 1.2 \times 6 = 7.2$  V [2]
- (ii) p.d. across Y and internal resistance  $r = 4.8$  (V) [ $12 - 7.2$ ]  
 resistance of  $Y + r = 4.8 / 1.2 = 4(\Omega)$   
 resistance of  $Y = 4 - 0.5 = 3.5 \Omega$   
 or  
 $R_{\text{total}} = 12 / 1.2 = 10 (\Omega)$   
 $X + r = 6.5 (\Omega)$   
 resistance of  $Y = 3.5 \Omega$  [3]
- (iii)  $P = I^2 r$   
 $= (1.2)^2 \times 0.5 = 0.72$  W [2]
- (c) terminal p.d. increases as  $R$  is increased  
 current decreases so there are less lost volts [1]

**54. O/N 14/P23/Q6**

**(a)**  $R = \rho l / A$

$$A = [\pi \times (0.38 \times 10^{-3})^2] / 4 \quad (= 0.113 \times 10^{-6} \text{ m}^2)$$

$$R = (4.5 \times 10^{-7} \times 1.00) / ([\pi \times (0.38 \times 10^{-3})^2] / 4) = 4.0 \text{ (3.97)} \Omega \quad [3]$$

**(b) (i)**  $I = V / R$

$$= 2.0 / 5.0 = 0.4(0) \text{ A} \quad [2]$$

**(ii)** p.d. across BD =  $4 \times 0.4 = 1.6 \text{ V}$  [1]

**(iii)** p.d. across BC ( $l$ ) =  $1.5 \text{ (V)}$

$$\text{BC } (l) = (1.5 / 1.6) \times 100 = 94 \text{ (93.75) cm} \quad [2]$$

**(c)** p.d. across wire not balancing e.m.f. of cell OR cell Y has current energy lost or lost volts due to internal resistance [2]